MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE BHL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital multiplexer module.

1.2 Classification. This specification covers one class of modules

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.970 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BHL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/66-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

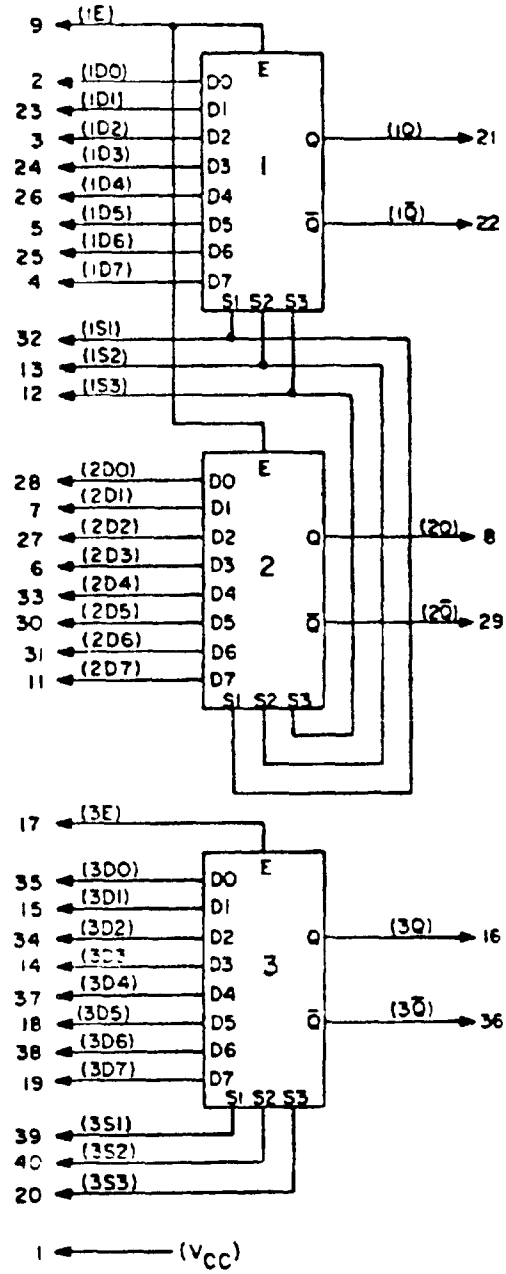
3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

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NOTES
1. 1D0-1D7, 2D0-2D7, 3D0-3D7 = Data inputs
2. 1E, 3E = Enable inputs
3. 1S1-1S3, 3S1-3S3 = Select inputs
4. V_{CC} = 5.0 V ±10 %
5. GND = 0 V REF

* FIGURE 1. BHL module function diagram, multiplexer.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B6	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	mV
C1-C28	Input breakdown current	V	S	0	100	0	100	0	100	0	100	µA
C29-C32	Input breakdown current	V	S	0	200	0	200	0	200	0	200	µA
D1-D6	Output short circuit current	VI	X	-6	-100	-6	-100	-6	-100	-6	-100	mA
E1-E6	Output leakage current	VII	X	0	250	0	250	0	250	0	250	µA
F1-F12	Output voltage, high level	VIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G1-G6	Output voltage, low level	VIII	X	0	500	0	500	0	500	0	500	mV
H1-H24	Functional	IX	X	Test vectors must provide output vectors as specified in table IX								
J1-J32	Input clamp diode voltage	X	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
K1, K10-K12	Input current, high level	XI	X	0	40	0	40	0	40	0	40	µA
K2-K9, K13-K32	Input current, high level	XI	X	0	20	0	20	0	20	0	20	µA
L1-L4	Input current, low level	XII	X	-200	-800	-200	-800	-200	-800	-200	-800	µA
L5-L20, L25-L32	Input current, low level	XII	X	-100	-440	-100	-440	-100	-440	-100	-440	µA
L21-L24	Input current, low level	XII	X	-100	-400	-100	-400	-100	-400	-100	-400	µA
M	Power supply current, low level	XIII	X	5	33	5	33	5	33	5	33	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1 N6	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	10	60	10	60	10	60	10	60	ns
N7 N12	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	10	130	10	130	10	130	10	130	ns
P1 P6	Transition time (t_{THL}), C_L = 50 pF	XIV	S	3	15	3	15	3	20	3	20	ns
P7 P12	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	6	50	6	50	6	55	6	55	ns
Q1-Q24	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	4	22	4	22	4	25	4	25	ns
Q25-Q33	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	10	35	10	35	10	38	10	38	ns
Q34-Q36	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	8	33	8	33	8	36	8	36	ns
Q37-Q60	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	8	36	8	36	8	39	8	39	ns
Q61-Q69	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	48	15	48	15	52	15	52	ns
Q70-Q72	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	44	15	44	15	47	15	47	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q73-Q96	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	10	28	10	28	10	31	10	31	ns
Q97-Q105	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	10	50	10	50	10	53	10	53	ns
Q106-Q108	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	8	34	8	34	8	38	8	38	ns
Q109-Q132	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	49	15	49	15	54	15	54	ns
Q133-Q141	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	20	64	20	64	20	71	20	71	ns
Q142-Q144	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	50	15	50	15	53	15	53	ns
Q145-Q153	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	8	30	8	30	8	33	8	33	ns
Q154-Q162	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	10	35	10	35	10	38	10	38	ns
Q163-Q171	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	41	15	41	15	44	15	44	ns
Q172-Q180	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	15	48	15	48	15	51	15	51	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1-R24	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	6	26	6	26	6	30	6	30	ns
R25-R33	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	10	45	10	45	10	50	10	50	ns
R34-R36	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	8	29	8	29	8	33	8	33	ns
R37-R60	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	10	40	10	40	10	44	10	44	ns
R61-R69	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	15	56	15	56	15	64	15	64	ns
R70-R72	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	15	43	15	43	15	48	15	48	ns
R73-R96	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	10	33	10	33	10	37	10	37	ns
R97-R105	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	15	51	15	51	15	56	15	56	ns
R106-R108	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	15	43	15	43	15	47	15	47	ns
R109-R132	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	15	48	15	48	15	54	15	54	ns

See footnotes at end of table

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R133-R141	Propagation delay time (t_{pLH}), C_L = 150 pF	XV	Q	20	62	20	62	20	68	20	68	ns
R142-R144	Propagation delay time (t_{pLH}), C_L = 150 pF	XV	Q	20	54	20	54	20	59	20	59	ns
R145-R153	Propagation delay time (t_{pLH}), C_L = 50 pF	XV	S	6	30	6	30	6	33	6	33	ns
R154-R162	Propagation delay time (t_{pLH}), C_L = 50 pF	XV	S	15	44	15	44	15	49	15	49	ns
R163-R171	Propagation delay time (t_{pLH}), C_L = 150 pF	XV	Q	15	43	15	43	15	48	15	48	ns
R172-R180	Propagation delay time (t_{pLH}), C_L = 150 pF	XV	Q	20	56	20	56	20	62	20	62	ns
S1-S6	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
T1-T6	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W28	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
W29-W32	Input capacitance	XX	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A,B,C,D, or E are part of a sequence of tests, the order of testing shall be as follows A,B,C,D,E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1 Component temperature test. The component temperature tests shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "A" in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/66-1 supersedes part number M28787/66. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5 V dc ±10 percent
Maximum power dissipation (V_{CC} = 5.5 V dc)	165 mW (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, 5.5 V maximum
Input current	
Low level (V_{IL} = 0.0 V)	
1D0-1D7, 2D0-2D7, 3D0-3D7	-0.44 mA maximum
3E, 3S1-3S3	-0.40 mA maximum
1E, 1S1-1S3	-0.80 mA maximum
Low level (V_{IL} = 0.4 V)	
1D0-1D7, 2D0-2D7, 3D0-3D7	-0.40 mA maximum
3E, 3S1-3S3	-0.36 mA maximum
1E, 1S1-1S3	-0.72 mA maximum
High level (V_{IH} = 2.5 V)	
1D0-1D7, 2D0-2D7, 3D0-3D7, 3E, 3S1-3S3	20 µA maximum
1E, 1S1-1S3	40 µA maximum
High level (V_{IH} = 5.5 V)	
1D0-1D7, 2D0-2D7, 3D0-3D7, 3E, 3S1-3S3	100 µA maximum
1E, 1S1-1S3	200 µA maximum
Output voltage	
Low level (I_{OL} = 6.0 mA)	0.5 V maximum
High level (I_{OH} = -400 µA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	6.0 mA
High level (V_{OH} = 2.5 V)	-400 µA

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Propagation delay time (t_{PH_L})

D to $\overline{Q}$, out-of-phase,	C_L = 50 pF	4 ns minimum, 25 ns maximum
	C_L = 150 pF	8 ns minimum 39 ns maximum
D to Q, in-phase,	C_L = 50 pF	10 ns minimum, 31 ns maximum
	C_L = 150 pF	15 ns minimum, 54 ns maximum
S to $\overline{Q}$, in-phase,	C_L = 50 pF	8 ns minimum, 33 ns maximum
	C_L = 150 pF	15 ns minimum, 44 ns maximum
S to $\overline{Q}$, out-of-phase,	C_L = 50 pF	10 ns minimum 38 ns maximum
	C_L = 150 pF	15 ns minimum, 52 ns maximum
S to Q, in-phase,	C_L = 50 pF	10 ns minimum, 53 ns maximum
	C_L = 150 pF	20 ns minimum, 71 ns maximum
S to Q, out-of-phase,	C_L = 50 pF	10 ns minimum, 38 ns maximum
	C_L = 150 pF	15 ns minimum, 51 ns maximum
E to $\overline{Q}$,	C_L = 50 pF	8 ns minimum, 35 ns maximum
	C_L = 150 pF	15 ns minimum, 47 ns maximum
E to Q,	C_L = 50 pF	8 ns minimum, 38 ns maximum
	C_L = 150 pF	15 ns minimum, 53 ns maximum

Propagation delay time (t_{PL_H})

D to $\overline{Q}$, out-of-phase,	C_L = 50 pF	6 ns minimum, 30 ns maximum
	C_L = 150 pF	15 ns minimum 44 ns maximum
D to Q, in-phase,	C_L = 50 pF	10 ns minimum, 37 ns maximum
	C_L = 150 pF	15 ns minimum 54 ns maximum
S to $\overline{Q}$, in-phase,	C_L = 50 pF	6 ns minimum, 32 ns maximum
	C_L = 150 pF	8 ns minimum 42 ns maximum
S to $\overline{Q}$, out-of-phase,	C_L = 50 pF	10 ns minimum 50 ns maximum
	C_L = 150 pF	15 ns minimum, 64 ns maximum
S to Q, in-phase,	C_L = 50 pF	15 ns minimum, 56 ns maximum
	C_L = 150 pF	20 ns minimum, 68 ns maximum
S to Q, out-of-phase,	C_L = 50 pF	15 ns minimum 49 ns maximum
	C_L = 150 pF	20 ns minimum, 62 ns maximum
E to $\overline{Q}$,	C_L = 50 pF	8 ns minimum 33 ns maximum
	C_L = 150 pF	15 ns minimum, 48 ns maximum
E to Q,	C_L = 50 pF	15 ns minimum 47 ns maximum
	C_L = 150 pF	20 ns minimum, 59 ns maximum

* 6.5 Module function. This module contains three circuits, two of which consist of two 8-way multiplexers with common select (S) and enable (E) lines, and one circuit which consists of one 8-way multiplexer with independent select (S) and enable (E) lines.

* 6.5.1 Input to output relationship. Each multiplexer has the input to output relationship of table II for positive logic.

* 6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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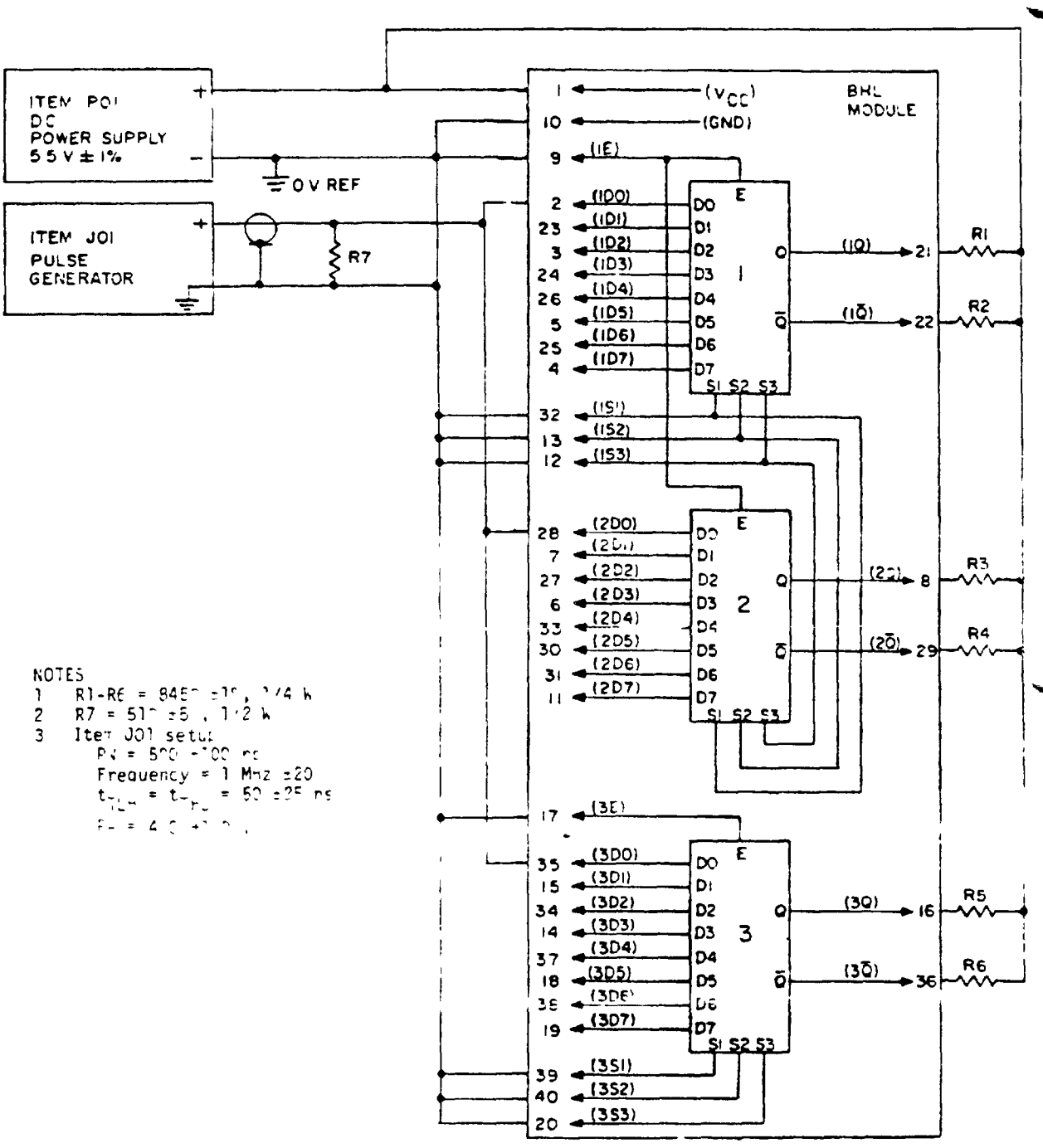
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* TABLE II. Input to output relationship. 1/-4/

Inputs												Output	
Select			Enable	Data input lines									
S3	S2	S1	E	D0	D1	D2	D3	D4	D5	D6	D7	Q	$\overline{Q}$
X	X	X	H	X	X	X	X	X	X	X	X	L	H
L	L	L	L	L	X	X	X	X	X	X	X	L	H
L	L	L	L	H	X	X	X	X	X	X	X	H	L
L	L	H	L	X	L	X	X	X	X	X	X	L	H
L	L	H	L	X	H	X	X	X	X	X	X	H	L
L	H	L	L	X	X	L	X	X	X	X	X	L	H
L	H	L	L	X	X	H	X	X	X	X	X	H	L
L	H	H	L	X	X	X	L	X	X	X	X	L	H
L	H	H	L	X	X	X	H	X	X	X	X	H	L
H	L	L	L	X	X	X	X	L	X	X	X	L	H
H	L	L	L	X	X	X	X	H	X	X	X	H	L
H	L	H	L	X	X	X	X	X	L	X	X	L	H
H	L	H	L	X	X	X	X	X	H	X	X	H	L
H	H	L	L	X	X	X	X	X	X	L	X	L	H
H	H	L	L	X	X	X	X	X	X	H	X	H	L
H	H	H	L	X	X	X	X	X	X	X	L	L	H
H	H	H	L	X	X	X	X	X	X	X	H	H	L

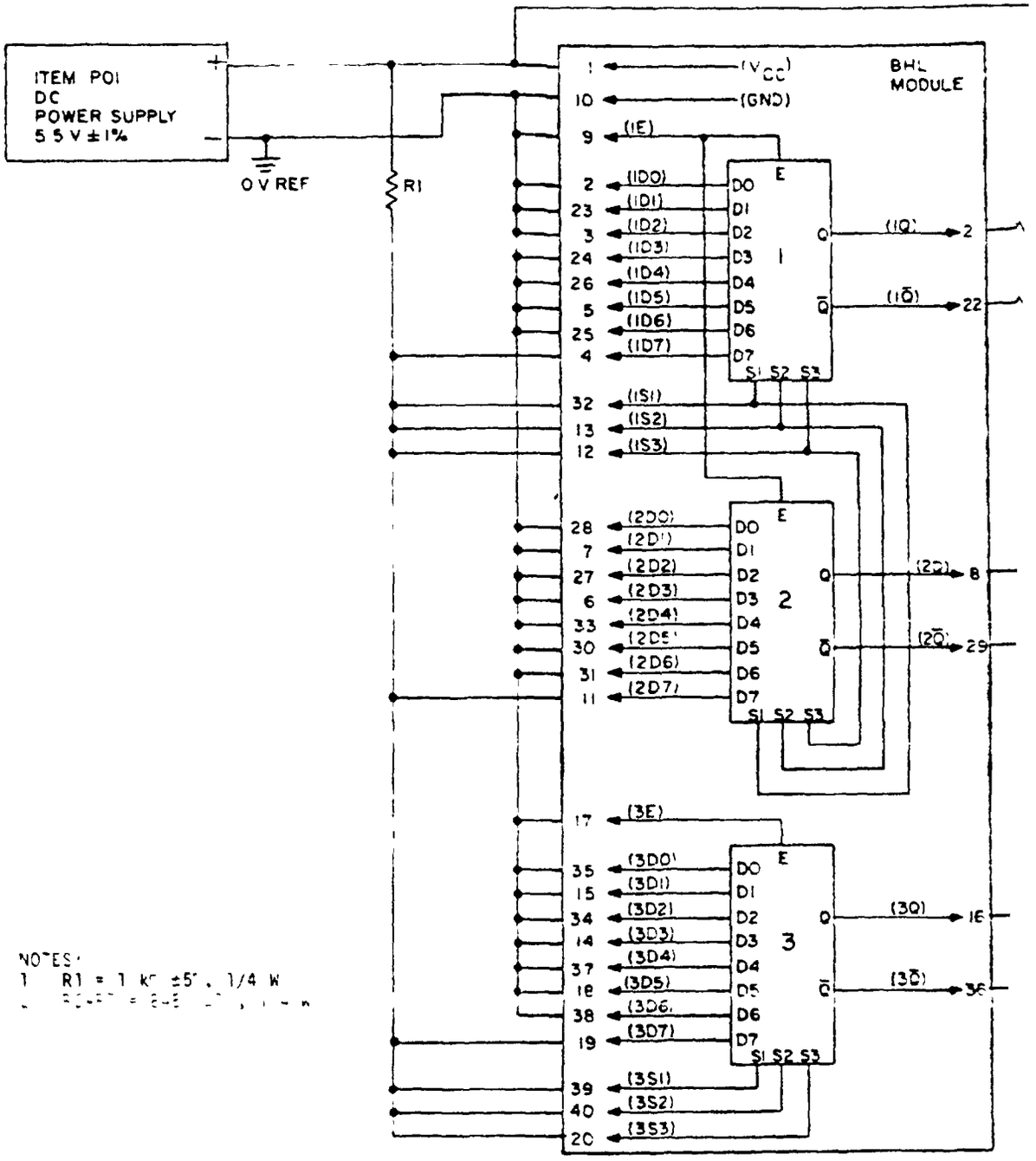
1/ $Q = \overline{E}(\overline{S1}\,\overline{S2}\,\overline{S3}D0 + S1\overline{S2}\,\overline{S3}D1 + \overline{S1}S2\overline{S3}D2 + S1S2\overline{S3}D3 + \overline{S1}\,\overline{S2}S3D4 + S1\overline{S2}S3D5 + \overline{S1}S2S3D6 + S1S2S3D7)$.
2/ X = Level does not affect output
3/ L = Low level.
4/ H = High level

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* FIGURE 2 Operating temperature and life test circuit

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* FIGURE 3 Component temperature test circuit

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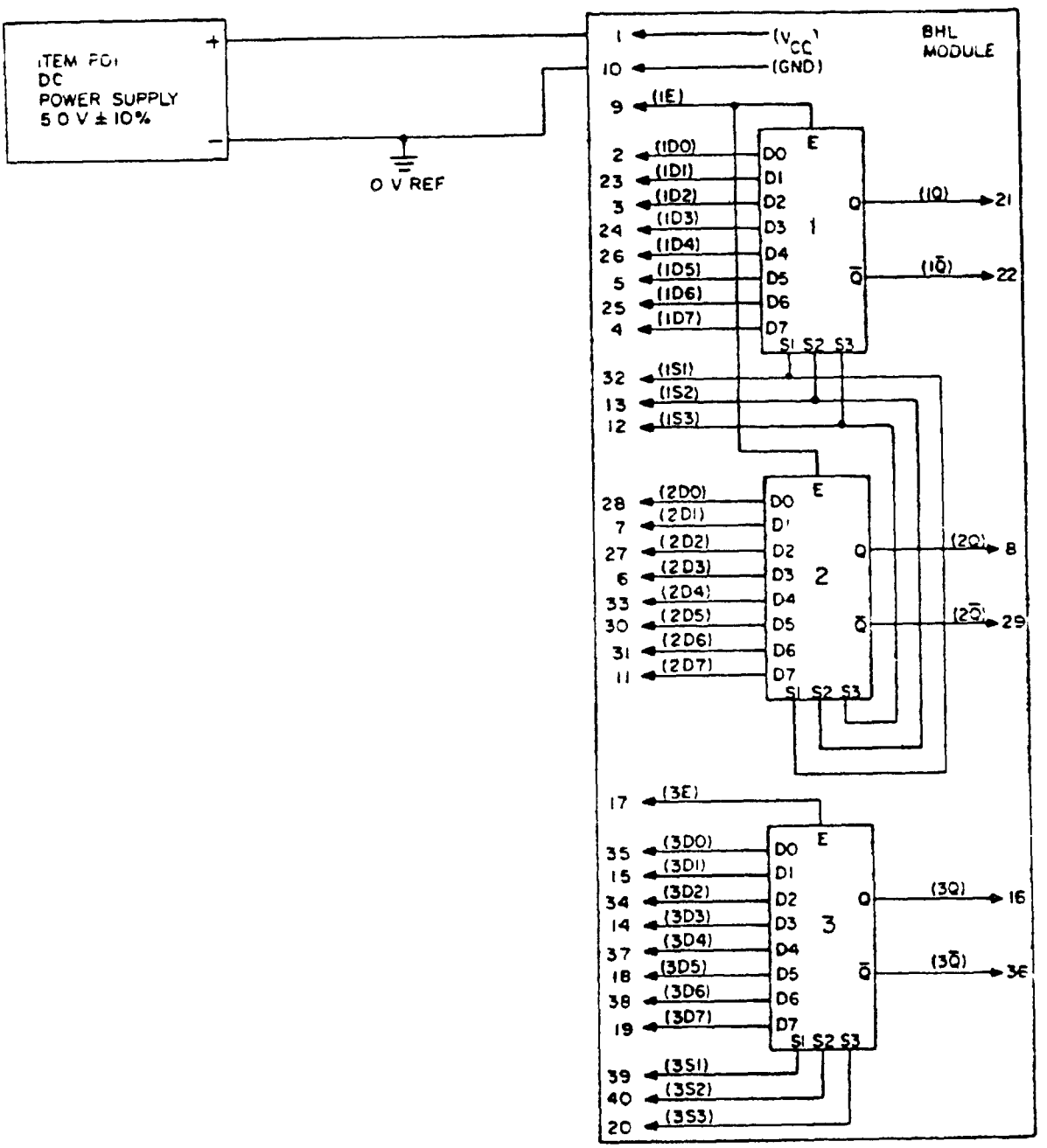


*FIGURE 4 Vibration test circuit

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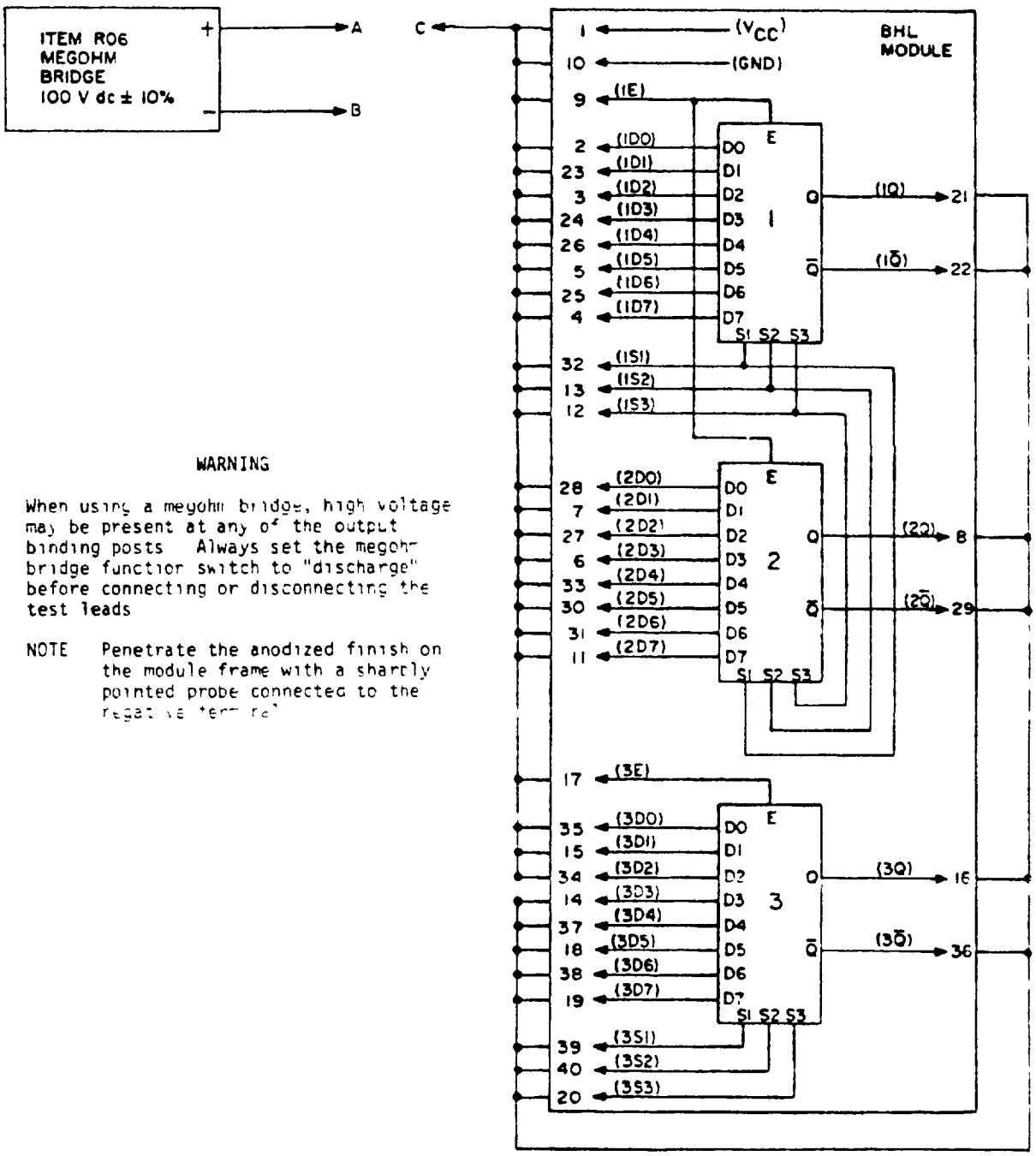
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	1Q V_{OUT} (low loaded)	9,17	Open	21	10	800	mV	6
B2	2Q	9,17	Open	8				
B3	3Q	9,17	Open	16				
B4	1$\overline{Q}$	32,13,24	12,9	22				
B5	2$\overline{Q}$	32,13,6	12,9	29				
B6	3$\overline{Q}$	39,40,14	20,17	36				

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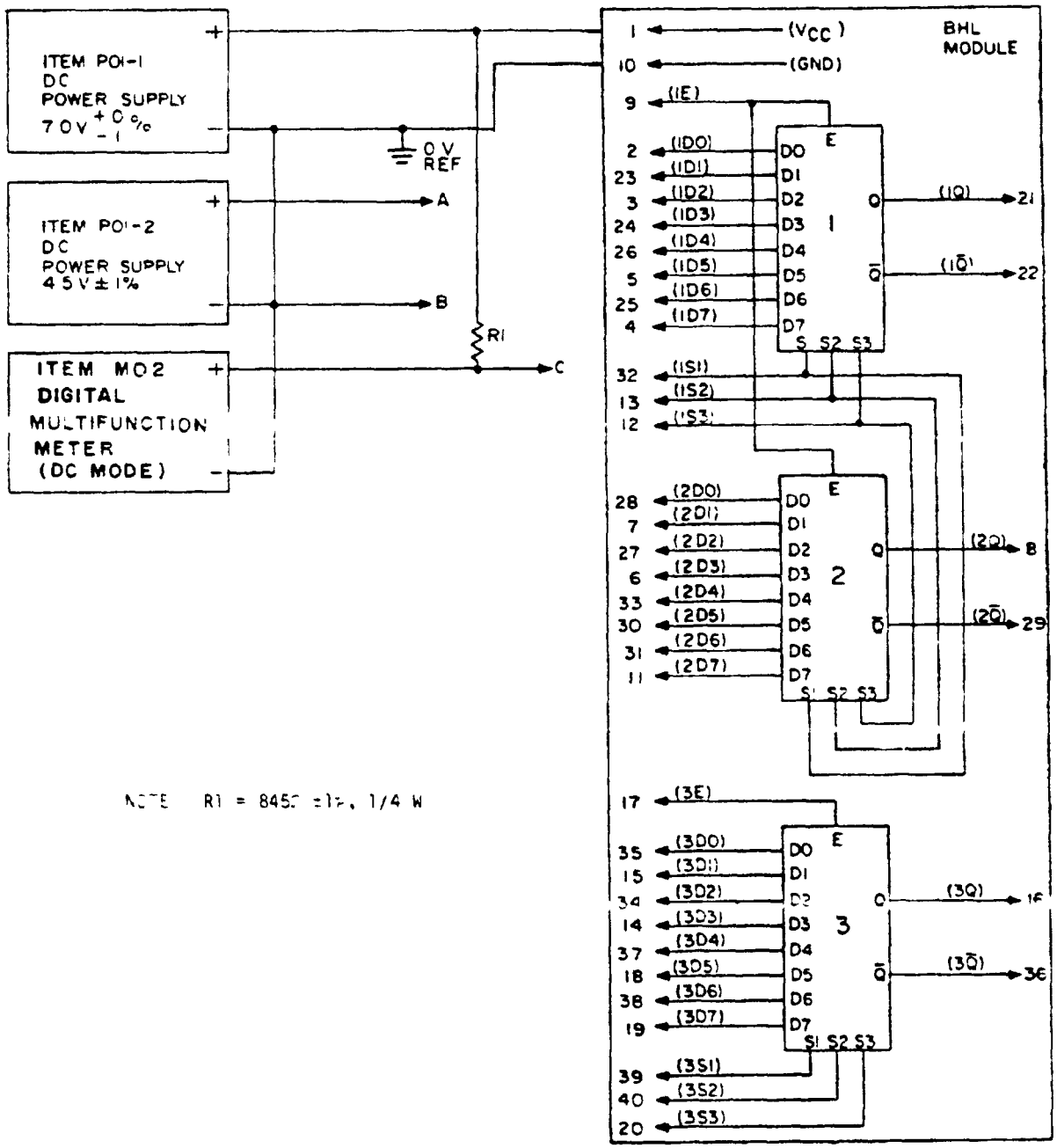
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 5 Isolation test circuit for table III

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NOTE R1 = 845Ω ±1%, 1/4 W

* FIGURE 6 V_{OL} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1D0 Input breakdown current	9,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,39, 40,20	2	0	100	µA	7
C2	2D0	9,2,23,3,24,26,5,25,4,32,13, 12,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,39,40, 20	28				
C3	1D1	9,2,3,24,26,5,25,4,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	23				
C4	2D1	9,2,23,3,24,26,5,25,4,32,13, 12,28,27,6,33,30,31,11,17, 35,15,34,14,37,18,19,40,20	7				
C5	1D2	9,2,23,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,39, 40,20	3				
C6	2D2	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,39,40,20	27				
C7	1D3	9,2,23,3,26,5,25,4,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	24				
C8	2D3	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,33,30,31,11,17, 35,15,34,14,37,18,38,19,40, 20	6				
C9	1D4	9,2,23,3,24,5,25,4,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,39,40,20	26				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C10	2D4 Input breakdown current	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,30,31,11,17,35, 15,34,14,37,18,38,19,39,40,20	33	0	100	µA	7
C11	1D5	9,2,23,3,24,26,25,4,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	5				
C12	2D5	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,31,11,17,35, 15,34,14,37,18,38,19,40,20	30				
C13	1D6	9,2,23,3,24,26,5,4,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,39,40,20	25				
C14	2D6	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,11,17,35, 15,34,14,37,18,38,19,39,40,20	31				
C15	1D7	9,2,23,3,24,26,5,25,32,13,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	4				
C16	2D7	9,2,23,3,24,26,5,25,4,32,13, 12,28 7,27,6,33,30,31,17,35, 15,34,14,37,18,38,19,40,20	11				
C17	3D0	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 15,34,14,37,18,38,19,39,40,20	35				
C18	3D1	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,34,14,37,18,38 19,39,40,20	15				

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TABLE V Input breakdown current test - Continued.

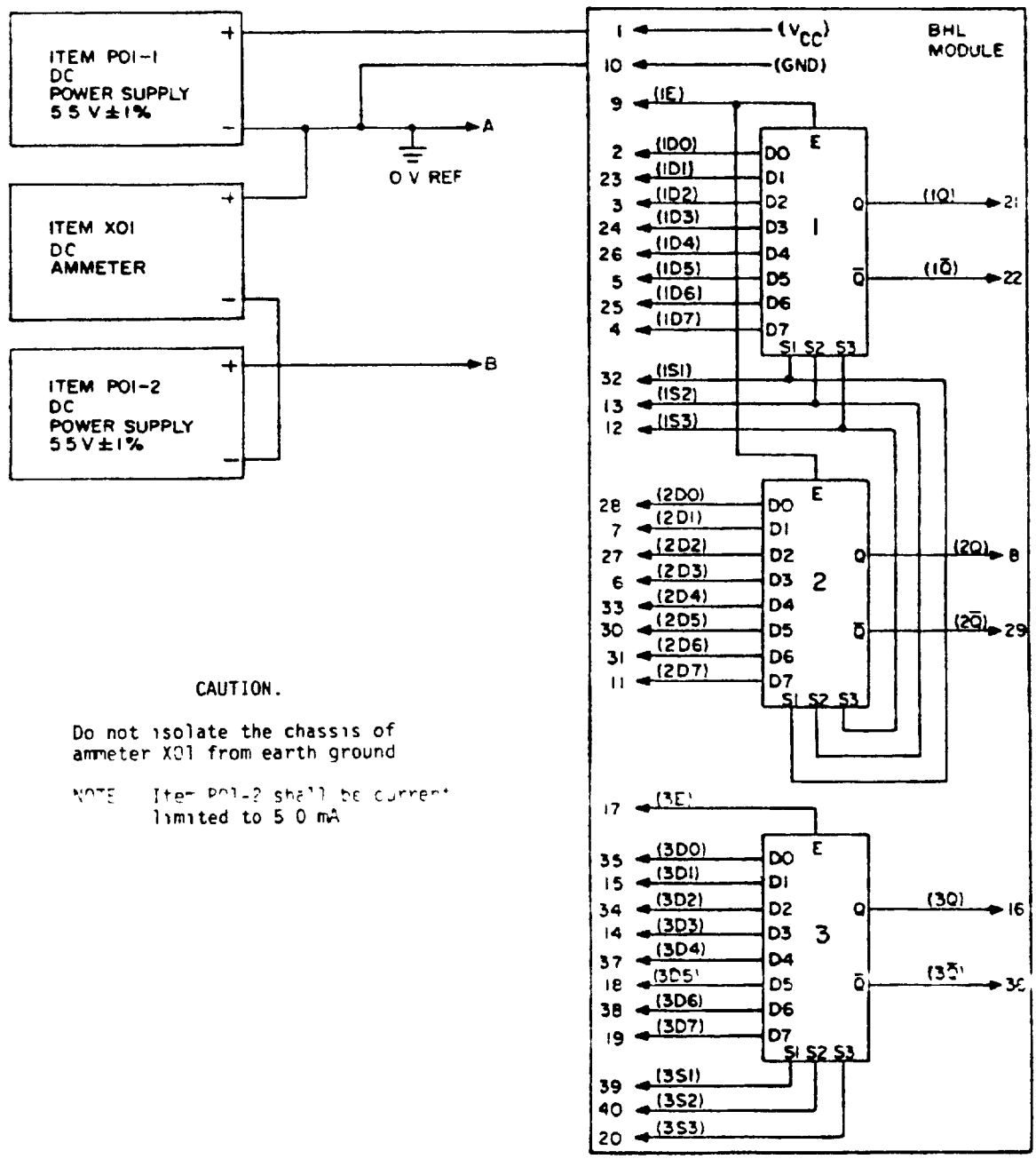
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	3D2 Input breakdown current	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,14,37,18,38,19,40,20	34	0	100	µA	7
C20	3D3	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,37,18,38,19,40,20	14				
C21	3D4	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,18,38,19,39,40,20	37				
C22	3D5	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,38,19,39,40,20	18				
C23	3D6	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,19,40,20	38				
C24	3D7	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,35, 15,34,14,37,18,38,17,40,20	19				
C25	3E	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,35, 15,34,14,37,18,38,19,39,40,20	17				
C26	3S1	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,40,20	39				
C27	3S2	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,20	40				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C28	3S3 Input breakdown current	9,2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,40	20	0	100	µA	7
C29	1E	2,23,3,24,26,5,25,4,32,13, 12,28,7,27,6,33,30,31,11, 17,35,15,34,14,37,18,38,19, 39,40,20	9		200		
C30	1S1	9,2,23,3,24,26,5,25,4,13, 12,28,7,27,6,33,30,31,11,17, 35,15,34,14,37,18,38,19,39, 40,20	32				
C31	1S2	9,2,23,3,24,26,5,25,4,32,12, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	13				
C32	1S3	9,2,23,3,24,26,5,25,4,32,13, 28,7,27,6,33,30,31,11,17,35, 15,34,14,37,18,38,19,40,20	12				

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* FIGURE 7 Input breakdown current test circuit for table V

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TABLE VI. Output short circuit current test.

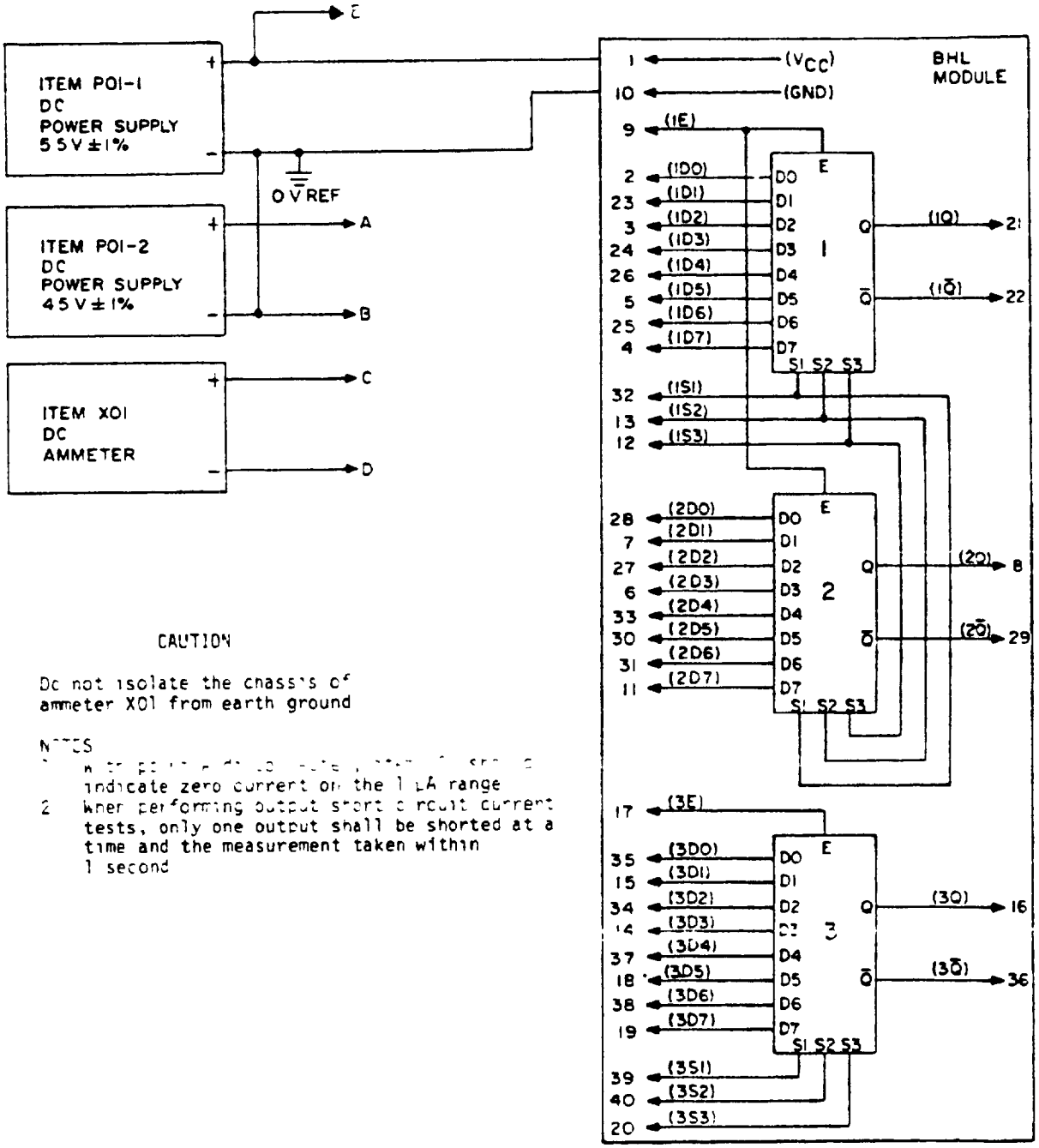
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	$1\bar{Q}$ Output short circuit current	9	2,23,3,24,26,5,25, 4,32,13,12,28,7,27, 6,33,30,31,11,17, 35,15,34,14,37,18, 38,19,39,40,20	8	22	-6	-100	mA	8
D2	$2\bar{Q}$	9	2,23,3,24,26,5,25, 4,32,13,12,28,7,27, 6,33,30,31,11,17, 35,15,34,14,37,18, 38,19,39,40,20		29				
D3	$3\bar{Q}$	17	2,23,3,24,26,5,25, 4,32,13,12,28,7,27, 6,33,30,31,11,9,35, 15,34,14,37,18,38, 19,39,40,20		36				
D4	1Q	12,13, 25	2,23,3,24,26,5,4, 32,28,7,9,27,6,33, 30,31,11,17,35,15, 34,14,37,18,38,19, 39,40,20		21				
D5	2Q	12,13, 31	2,23,3,24,26,5,25, 4,32,28,7,9,27,6, 33,30,11,17,35,15, 34,14,37,18,38,19, 39,40,20		8				
D6	3Q	20,40, 38	2,23,3,24,26,5,25, 4,32,28,7,9,27,6, 33,30,11,17,35,15, 34,14,37,18,19,39, 12,13		16				

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* TABLE VII. Output leakage current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	$1\bar{Q}$ Output leakage current	9	Open	E	22	0	250	µA	8
E2	$2\bar{Q}$	9	Open		29				
E3	$3\bar{Q}$	17	Open		36				
E4	1Q	12,13,25	9,32		21				
E5	2Q	12,13,31	9,32		8				
E6	3Q	20,40,38	17,39		16				

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CAUTION

Do not isolate the chassis of ammeter XOI from earth ground

NOTES

1. [illegible] indicate zero current on the 1 µA range
2. When performing output short circuit current tests, only one output shall be shorted at a time and the measurement taken within 1 second

* FIGURE 8 Output short circuit current and output leakage current test circuit for tables VI and VII

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* TABLE VIII. Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1Q	Output voltage, high level	2	9,12,13,32,20,40, 39,17,35,28	21	2.5	4.5	V	9
F2	1Q		2,28,35	9,12,13,32,20,40, 39,17	21				
F3	1$\overline{Q}$		Open	9,12,13,32,20,40, 39,17,2,28,35	22				
F4	1$\overline{Q}$		28.35	9,12,13,32,20,40, 39,17,2	22				
F5	2Q		28	9,12,13,32,20,40, 39,17,35,2	8				
F6	2Q		28,35,2	9,12,13,32,20,40, 39,17	8				
F7	2$\overline{Q}$		Open	9,12,13,32,20,40, 39,17,2,28,35	29				
F8	2$\overline{Q}$		2,35	9,12,13,32,20,40, 39,17,28	29				
F9	3Q		35	9,12,13,32.20,40, 39,17,2,28	16				
F10	3Q		35,28.2	9,12,13,32,20,40, 39,17	16				
F11	3$\overline{Q}$		Open	9,12,13,32,20,40, 39,17,2,28,35	36				
F12	3$\overline{Q}$		2,28	9,12,13,32,20,40, 39,17,35	36				

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* TABLE VIII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	Q1	Output voltage, low level	Open	9,17,13,32,2	21	0	500	mV	9
G2	Q2	↓	Open	9,12,13,32,28	8	↓	↓	↓	↓
G3	Q3		Open	17,20,39,40,35	16				
G4	$Q\overline{1}$		2	9,12,13,32	22				
G5	$Q\overline{2}$		28	9,12,13,32	29				
G6	$Q\overline{3}$		35	17,20,39,40	36				

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* TABLE IX. Functional test.

Test condition	Parameter	Connect			Output states						Initial limits 25° ±5°C ambient		Test figure
		A to pin	B to pin	C to pin	1Q	2Q	3Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	Min	Max	
H1	Functional	2,3,4,5,6,7,9,11, 14,15,17,18,19,23, 24,25,26,27,29,30, 31,33,34,35,37,38	12,13,20,32,39,40	2/ 3/	L	L	L	H	H	H	1/		9
H2		2,3,4,5,6,7,9,11, 14,15,17,18,19,23, 24,25,26,27,28,30, 31,32,33,34,36,37, 38,39	12,13,20,40										
H3		2,3,4,5,6,7,9,11, 13,14,15,17,18,19, 23,24,25,26,27,28, 30,31,33,34,36,37, 38,40	12,20,32,39										
H4		2,3,4,5,6,7,9,11, 13,14,15,17,18,19, 23,24,25,26,27,28, 30,31,32,33,34,35, 37,38,39,40	12,20										
H5		2,3,4,5,6,7,9,11, 12,14,15,17,18,19, 20,23,24,25,26,27, 28,30,31,33,34,35, 37,38	13,32,39,40										
H6		2,3,4,5,6,7,9,11, 12,14,15,17,18,19, 20,23,24,25,26,27, 28,30,31,32,33,34, 35,37,38,39	13,40										
H7		2,3,4,5,6,7,9,11, 12,13,14,15,17,18, 19,20,23,24,25,26, 27,28,30,31,33,34, 35,37,38,40	32,39										

e footnotes at end of table.

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* TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect			Output states						Initial limits 25°±5°C ambient		Test figure
		A to pin	B to pin	C to pin	1Q	2Q	3Q	1$\bar{Q}$	2$\bar{Q}$	3$\bar{Q}$	Min	Max	
HI8	Functional	9,17,12,20,13,40,32, 39,2,28,35,23,7,15, 3,27,34,24,6,14,26, 33,37,5,30,18,25,31, 38,4,11,19	Open	2/ 3/	L	L	L	H	H	H	1/		9
HI9		23,7,15,3,27,34,24, 6,14,26,33,37,5,30, 18,25,31,38,4,11,19	9,17,12,20,13,40, 32,39,2,28,35		L	L	L	H	H	H			
HI10		2,28,35	9,17,12,20,13,40, 32,39,23,7,15,3, 27,34,24,6,14,26, 33,37,5,30,18,25, 31,38,4,11,19		H	H	H	L	L	L			
HI11		32,39,2,28,35,3,27, 34,24,6,14,26,33, 37,5,30,18,25,31, 38,4,11,19	9,17,12,20,13,40, 23,7,15		L	L	L	H	H	H			
HI12		32,39,23,7,15	9,17,12,20,13,40,2, 28,35,3,27,34,24,6, 14,26,33,37,5,30, 18,25,31,38,4,11,19		H	H	H	L	L	L			
HI13		2,28,35,23,7,15,24, 6,14,26,33,37,5,30, 18,25,31,38,4,11, 19,13,40	9,17,12,20,32,39,3, 27,34		L	L	L	H	H	H			
HI14		13,40,3,27,34	9,17,12,20,32,39,2, 28,35,23,7,15,24,6, 14,26,33,37,5,30, 18,25,31,38,4,11,19		H	H	H	L	L	L			

See footnotes at end of table.

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* TABLE IX. Functional test - Continued

Test condition	Parameter	Connect			Output states						Initial limits 25° ±5°C ambient		Test figure
		A to pin	B to pin	C to pin	1Q	2Q	3Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	Min	Max	
H15	Functional	13,40,32,39,2,28, 35,23,7,15,3,27,34, 26,33,37,5,30,18, 25,31,38,4,11,19	9,17,12,20,24,6,14	2/ 3/	L	L	L	H	H	H	1/		9
H16		13,40,32,39,24,6,14	9,17,12,20,2,28,35, 23,7,15,3,27,34,26, 33,37,5,30,18,25, 31,38,4,11,19		H	H	H	L	L	L			
H17		12,20,2,28,35,23,7, 15,3,27,34,24,6,14, 5,30,18,25,31,38,4, 11,19	9,17,13,40,32,39, 26,33,37		L	L	L	H	H	H			
H18		12,20,26,33,37	9,17,13,40,32,39,2, 28,35,23,7,15,3,27, 34,24,6,14,5,30,18, 25,31,38,4,11,19		H	H	H	L	L	L			
H19		12,20,32,39,2,28, 35,23,7,15,3,27,34, 24,6,14,26,33,37, 25,31,38,4,11,19	9,17,13,40,5,30,18		L	L	L	H	H	H			
H20		12,20,32,39,5,30,18	9,17,13,40,2,28,35, 23,7,15,3,27,34,24, 6,14,26,33,37,25, 31,38,4,11,19		H	H	H	L	L	L			
H21		12,20,13,40,2,28, 35,23,7,15,3,27,34, 24,6,14,26,33,37,5, 30,18,4,11,19	9,17,32,39,25,31,38		L	L	L	H	H	H			

See footnotes at end of table.

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* TABLE IX. Functional test - Continued.

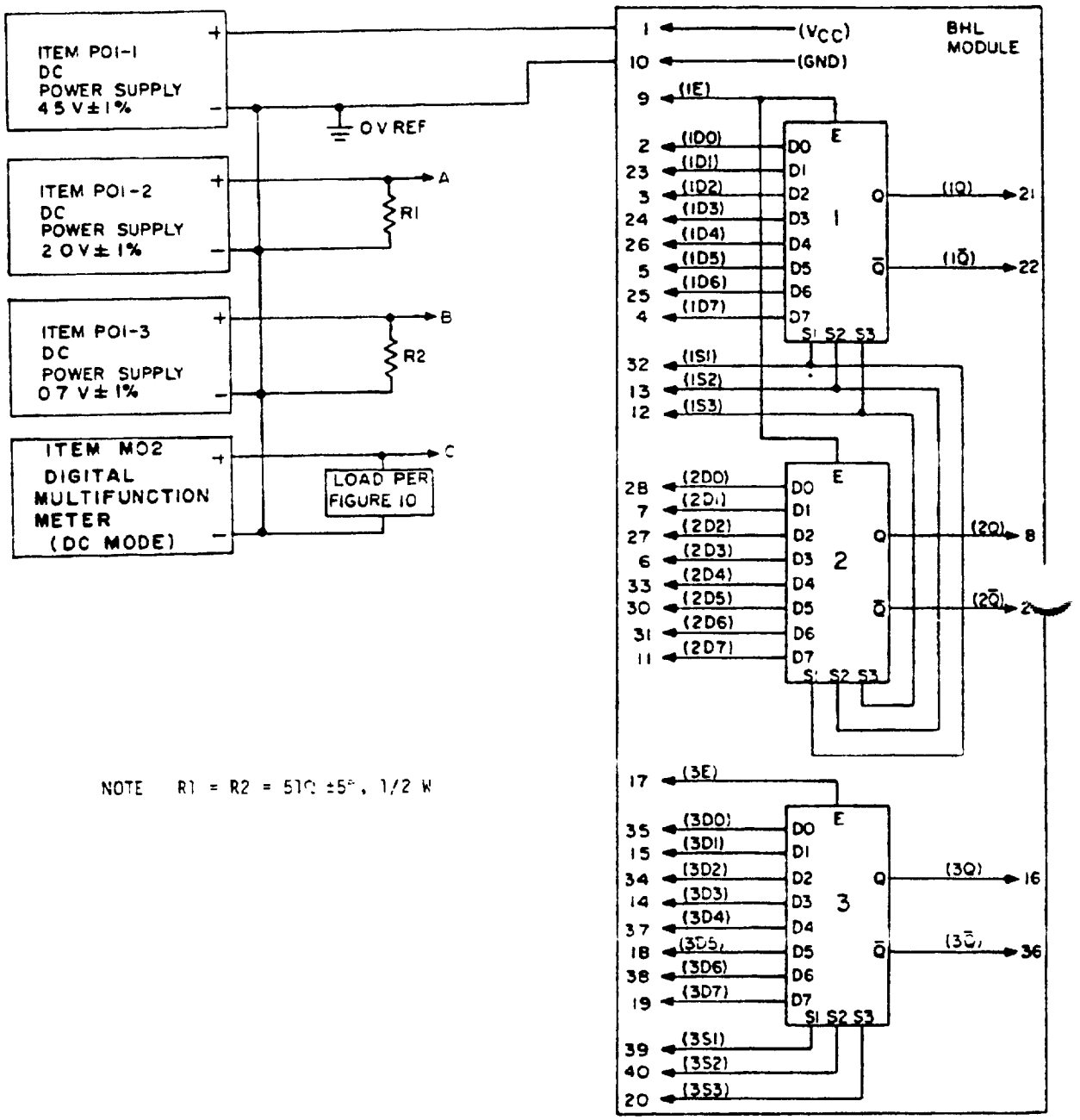
Test condition	Parameter	Connect			Output states						Initial limits 25° ±5°C ambient		Test figure
		A to pin	B to pin	C to pin	1Q	2Q	3Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	Min	Max	
H22	Functional	12,20,13,40,25,31, 38	9,17,32,39,2,28,35, 23,7,15,3,27,34,24, 6,14,26,33,37,5,30, 18,4,11,19	2/ 3/	H	H	H	L	L	L	1/		9
H23	Functional	12,20,13,40,32,39, 2,28,35,23,7,15,3, 27,34,24,6,14,26, 33,37,5,30,18,25, 31,38	9,17,4,11,19	2/ 3/	L	L	L	H	H	H	1/		9
H24	Functional	12,20,13,40,32,39, 4,11,19	9,17,2,28,35,23,7, 15,3,27,34,24,6,14, 26,33,37,5,30,18, 25,31,38	2/ 3/	H	H	H	L	L	L	1/		9

1/ Output logic states
L = Low level = 0.0 to 0.5 V.
H = High level = 2.5 to 4.5 V

2/ For test conditions H1-H24 verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output in succession after input conditions have been accomplished.

3/ 1Q, 2Q, 3Q, 1$\overline{Q}$, 2$\overline{Q}$, and 3$\overline{Q}$ output logic states are measured at output pins 21, 8, 16, 22, 29, and 36 respectively.

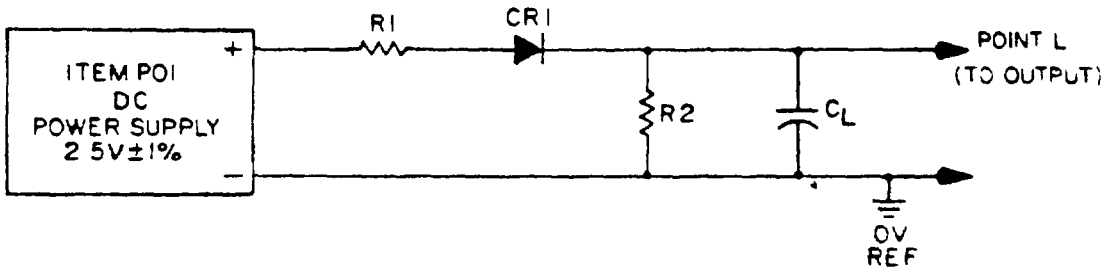
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NOTE R1 = R2 = 510 ±5%, 1/2 W

* FIGURE 9. Output voltage (high-level and low-level) and functional test circuit for tables I, II, and III

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NOTES:
1 CR1 = 1N3600, 1N4150, or equivalent
2 R1 = Select value for a current flow of 6 mA ±5 out of point L with point L held at 0.5 V.
3 R2 = 6.34 k ±1 , 1 t W
4 Test type S
C_L = 50 ±5 pF (including probe and parasitic capacitance)
Test type Q
C_L = 150 ±5 pF (including probe and parasitic capacitance) except for output crosstalk test which shall use 50 ±5 pF
5 Capacitor C_L is not required for dc electrical test conditions
6 Lead length from load circuit to module output for ac test shall be less than 1 inch (25.4 mm)

* FIGURE 10 Bus logic load circuit

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TABLE X. Input clamp diode voltage test.

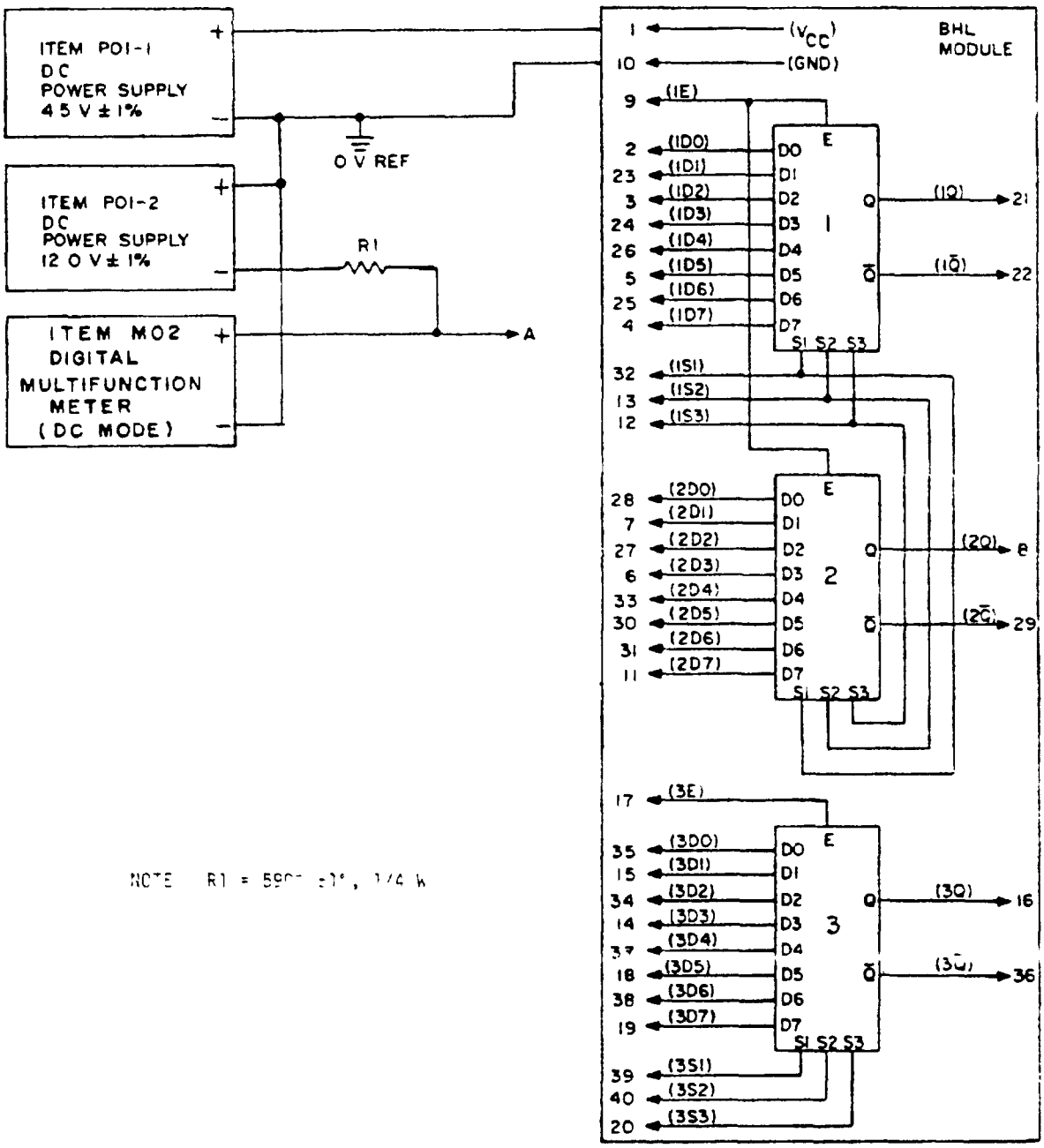
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	1E Input clamp diode voltage	9	-0.4	-1.5	V	11
J2	1D0	2				
J3	1D1	23				
J4	1D2	3				
J5	1D3	24				
J6	1D4	26				
J7	1D5	5				
J8	1D6	25				
J9	1D7	4				
J10	1S1	32				
J11	1S2	13				
J12	1S3	12				
J13	2D0	28				
J14	2D1	7				
J15	2D2	27				
J16	2D3	6				
J17	2D4	33				
J18	2D5	30				
J19	2D6	31				

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TABLE X. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect: A to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
J20	2D7 Input clamp diode voltage	11	-0.4	-1.5	V	11
J21	3E	17				
J22	3D0	35				
J23	3D1	15				
J24	3D2	34				
J25	3D3	14				
J26	3D4	37				
J27	3D5	18				
J28	3D6	38				
J29	3D7	19				
J30	3S1	39				
J31	3S2	40				
J32	3S3	20				

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NOTE R1 = 590 ±1%, 1/4 W

*FIGURE 11 Input clamp diode voltage test circuit for table X.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	1E	Input current, high level	2,3,4,5,6,7,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	9	0	40	uA	12
K2	1D0		3,4,5,6,7,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	2		20		
K3	1D1		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	23				
K4	1D2		2,4,5,6,7,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	3				
K5	1D3		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,35,37,38,39,40	24				
K6	1D4		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,27,28,30, 31,32,33,34,35,37,38,39,40	26				
K7	1D5		2,3,4,6,7,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	5				
K8	1D6		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,26,27,28,30, 31,32,33,34,35,37,38,39,40	25				
K9	1D7		2,3,5,6,7,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	4				

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TABLE XI Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K10	1S1	Input current, high level	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,33,34,35,37,38,39,40	32	0	40	µA	12
K11	1S2		2,3,4,5,6,7,9,11,12,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	13		40		
K12	1S3		2,3,4,5,6,7,9,11,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	12		40		
K13	2D0		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,30, 31,32,33,34,35,37,38,39,40	28		20		
K14	2D1		2,3,4,5,6,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	7				
K15	2D2		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,28,30, 31,32,33,34,35,37,38,39,40	27				
K16	2D3		2,3,4,5,7,9,11,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	6				
K17	2D4		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,34,35,37,38,39,40	33				
K18	2D5		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 31,32,33,34,35,37,38,39,40	30				

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K19	2D6	Input current, high level	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,32,33,34,35,37,38,39,40	31	0	20	μA	12
K20	2D7		2,3,4,5,6,7,9,12,13,14,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	11				
K21	3E		2,3,4,5,6,7,9,11,12,13,14,15, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	17				
K22	3D0		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,37,38,39,40	35				
K23	3D1		2,3,4,5,6,7,9,11,12,13,14,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	15				
K24	3D2		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,35,37,38,39,40	34				
K25	3D3		2,3,4,5,6,7,9,11,12,13,15,17, 18,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	14				
K26	3D4		2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,35,38,39,40	37				
K27	3D5		2,3,4,5,6,7,9,11,12,13,14,15, 17,19,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	18				

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K28	3D6	Input current, high level	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,35,37,39,40	38	0	20	µA	12
K29	3D7	↓	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,20,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	19	↓	↓	↓	↓
K30	3S1	↓	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,35,37,38,40	39	↓	↓	↓	↓
K31	3S2	↓	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,20,23,24,25,26,27,28, 30,31,32,33,34,35,37,38,39	40	↓	↓	↓	↓
K32	3S3	↓	2,3,4,5,6,7,9,11,12,13,14,15, 17,18,19,23,24,25,26,27,28,30, 31,32,33,34,35,37,38,39,40	20	↓	↓	↓	↓

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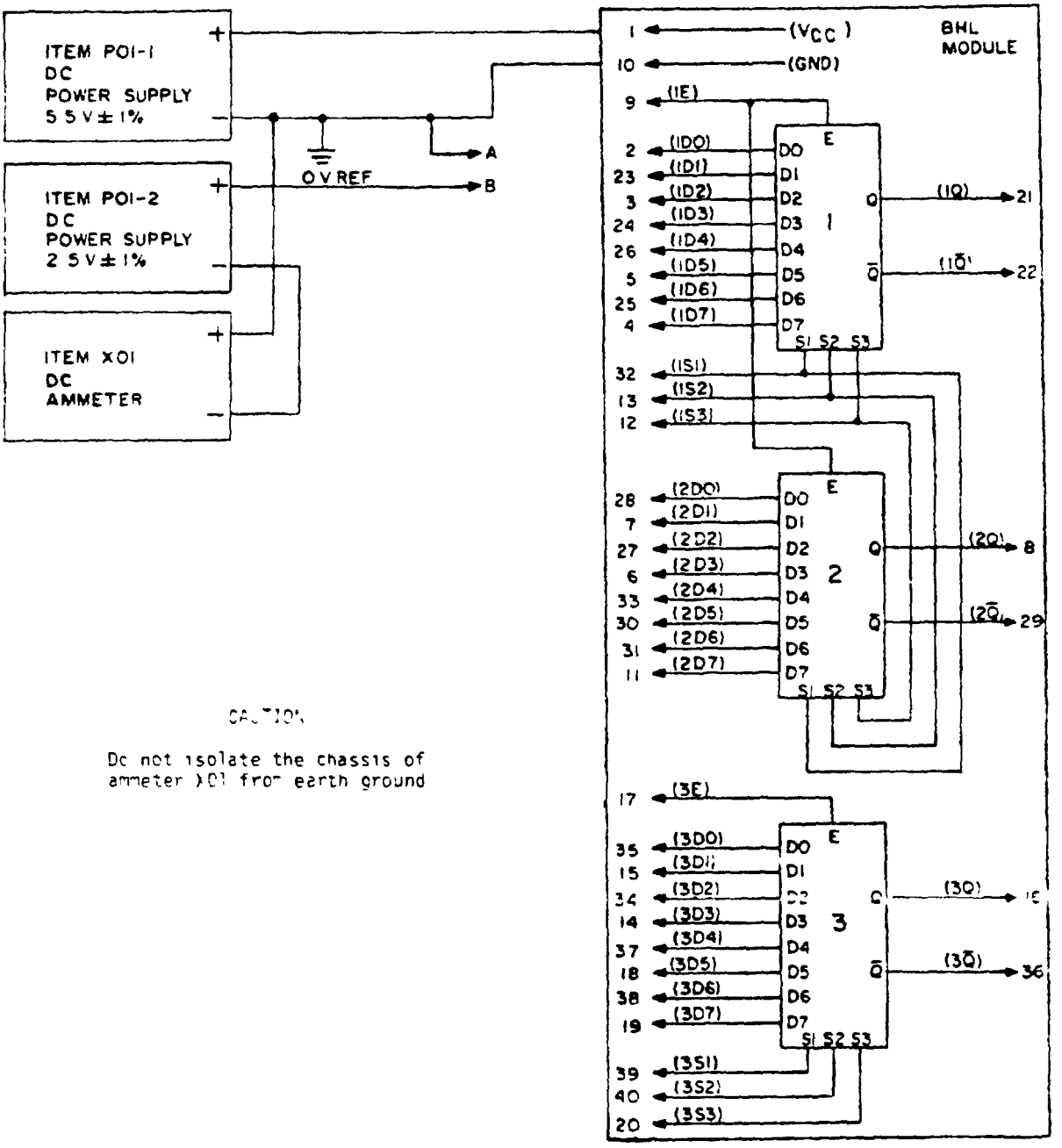


*FIGURE 12 Input current (high-level) test circuit for table XI

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* TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1E Input current, low level	Open	Open	9	-200	-800	µA	13
L2	1S1			32				
L3	1S2			13				
L4	1S3			12				
L5	1D0		32,13,12,9	2	-100	-440		
L6	1D1	32	12,13,9	23				
L7	1D2	13	32,12,9	3				
L8	1D3	32,13	12,9	24				
L9	1D4	12	32,13,9	26				
L10	1D5	32,12	13,9	5				
L11	1D6	12,13	32,9	25				
L12	1D7	32,12,13	9	4				
L13	2D0	Open	32,13,12,9	28				
L14	2D1	32	12,13,9	7				
L15	2D2	13	32,12,9	27				
L16	2D3	32,13	12,9	6				
L17	2D4	12	32,13,9	33				
L18	2D5	32,12	13,9	30				
L19	2D6	12,13	32,9	31				

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* TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L20	2D7	Input current, low level	32,12,13	9	11	-100	-440	µA	13
L21	3E		Open	Open	17		-400		
L22	3S1				39				
L23	3S2				40				
L24	3S3				20				
L25	3D0			39,40,20 17	35		-440		
L26	3D1		39	20,40,17	15				
L27	3D2		40	39,20,17	34				
L28	3D3		39,40	20,17	14				
L29	3D4		20	17,39,40	37				
L30	3D5		39,20	17,40	18				
L31	3D6		20,40	39,17	38				
L32	3D7		39,40,20	17	19				

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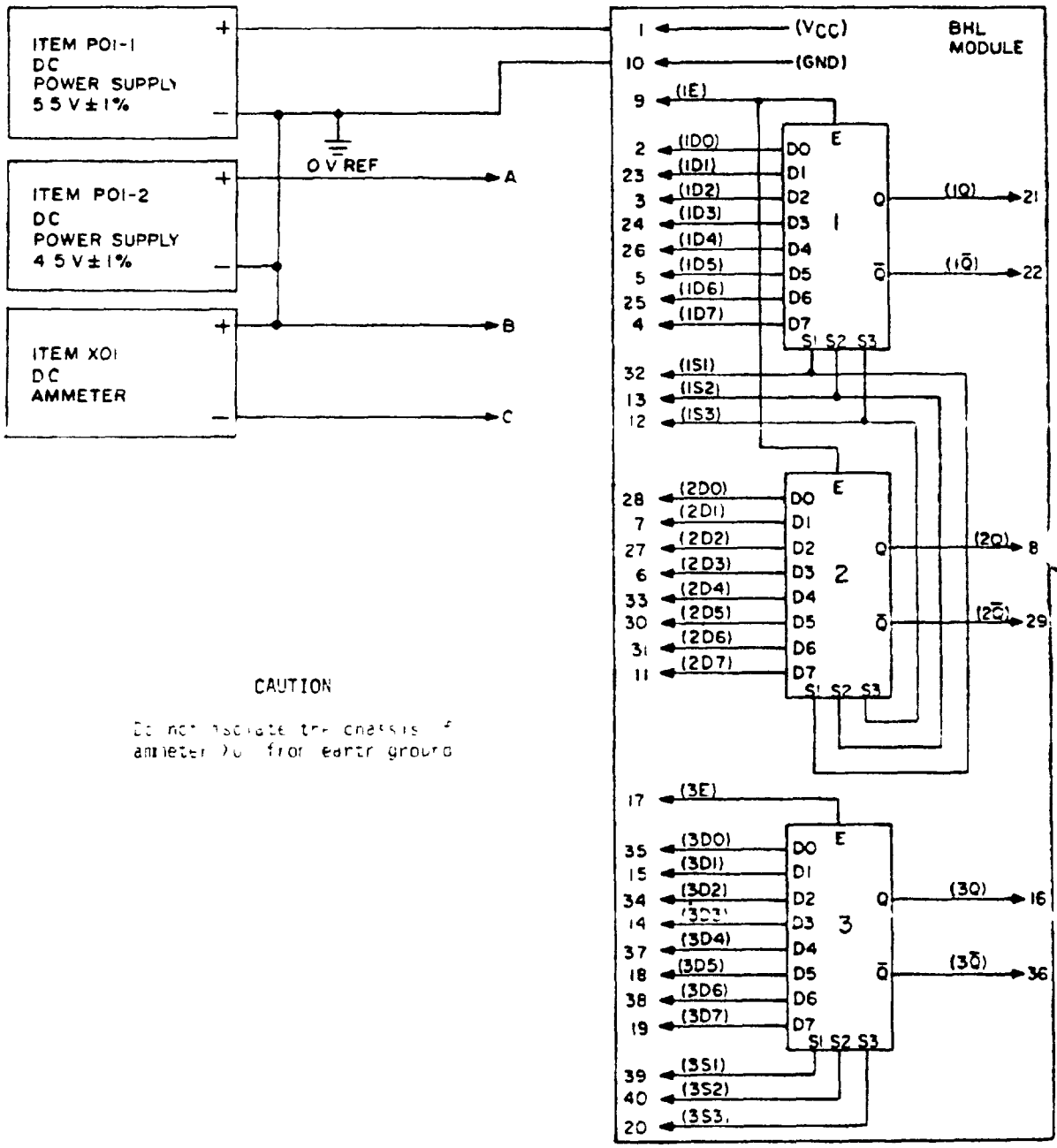


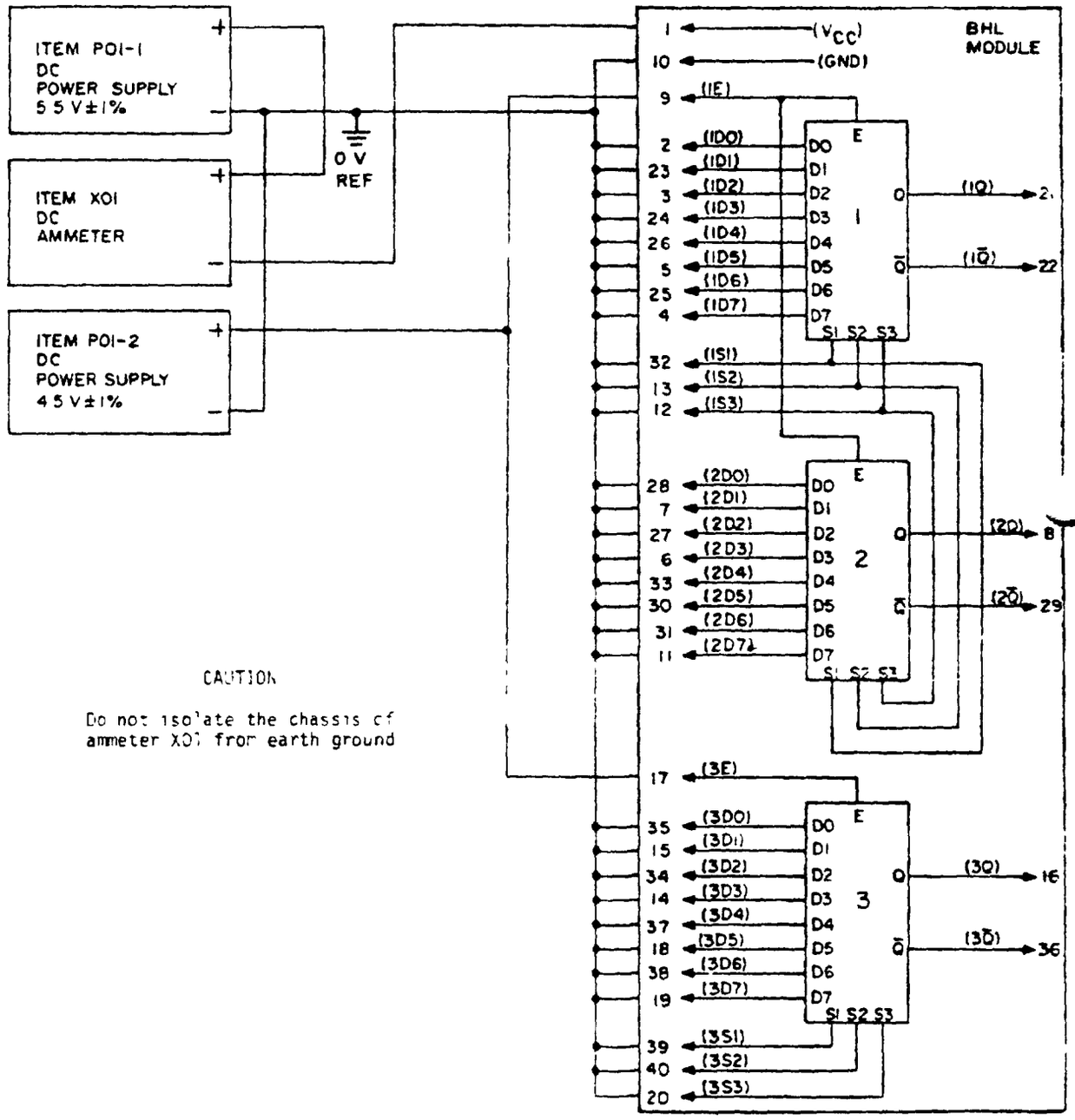
*FIGURE 13 Input current (low-level) test circuit for table XII

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* TABLE XIII. Power supply current (low-level) test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current, low level	5	33	mA	14

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* FIGURE 14. Power supply current (low-level) test circuit for table VIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Q	Transition time (t_{TLH}), C_L 50 pF	32,13	9,12	24	21	10	60	ns	15,16
N2	1$\overline{Q}$				24	22				15,17
N3	2Q				6	8				15,16
N4	2$\overline{Q}$				6	29				15,17
N5	3Q		39,40	17,20	14	16				15,16
N6	3$\overline{Q}$		39,40	17,20	14	36				15,17
N7	1Q	Transition time (t_{TLH}), C_L 150 pF	32,13	9,12	24	21		130		15,16
N8	1$\overline{Q}$				24	22				15,17
N9	2Q				6	8				15,16
N10	2$\overline{Q}$				6	29				15,17
N11	3Q		39,40	17,20	14	16				15,16
N12	3$\overline{Q}$		39,40	17,20	14	36				15,17
P1	1Q	Transition time (t_{THL}), C_L 50 pF	32,13	9,12	24	21	3	15		15,16
P2	1$\overline{Q}$				24	22				15,17
P3	2Q				6	8				15,16
P4	2$\overline{Q}$				6	29				15,17

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P5	3Q	Transition time (t_{THL}), C_L 50 pF	39,40	17,20	14	16	3	15	ns	15,16
P6	3$\overline{Q}$	Transition time (t_{THL}), C_L 50 pF	39,40	17,20	14	36	3	15		15,17
P7	1Q	Transition time (t_{THL}), C_L 150 pF	37,13	9,12	24	21	6	50		15,16
P8	1$\overline{Q}$				24	22				15,17
P9	2Q				6	8				15,16
P10	2$\overline{Q}$				6	29				15,17
P11	3Q		39,40	17,20	14	16				15,16
P12	3$\overline{Q}$		39,40	17,20	14	36				15,17

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1D0 to 1$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF	Open	9,12 13,32	2	22	4	22	ns	15,17
Q2	1D1 to 1$\overline{Q}$	32	9,13,12	23					
Q3	1D2 to 1$\overline{Q}$	13	9,12,32	3					
Q4	1D3 to 1$\overline{Q}$	32,13	9,12	24					
Q5	1D4 to 1$\overline{Q}$	12	9,32,13	26					
Q6	1D5 to 1$\overline{Q}$	12,32	9,13	5					
Q7	1D6 to 1$\overline{Q}$	13,12	9,32	25					
Q8	1D7 to 1$\overline{Q}$	12,13, 32	9	4					
Q9	2D0 to 2$\overline{Q}$	Open	9,12, 13,32	28	29				
Q10	2D1 to 2$\overline{Q}$	32	9,13,12	7					
Q11	2D2 to 2$\overline{Q}$	13	9,12,32	27					
Q12	2D3 to 2$\overline{Q}$	32,13	9,12	6					
Q13	2D4 to 2$\overline{Q}$	12	9,32,13	33					
Q14	2D5 to 2$\overline{Q}$	32,32	9,13	30					
Q15	2D6 to 2$\overline{Q}$	13,12	9,32	31					
Q16	2D7 to 2$\overline{Q}$	12,13, 32	9	11					

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q17	3D0 to 3$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 50 pF	Open	17,20, 39,40	35	36	4	22	ns	15,17
Q18	3D1 to 3$\overline{Q}$	39	17,20, 40	15					
Q19	3D2 to 3$\overline{Q}$	40	17,39, 20	34					
Q20	3D3 to 3$\overline{Q}$	39,40	17,20	14					
Q21	3D4 to 3$\overline{Q}$	20	17,39, 40	37					
Q22	3D5 to 3$\overline{Q}$	20,39	17,40	18					
Q23	3D6 to 3$\overline{Q}$	20,40	17,39	38					
Q24	3D7 to 3$\overline{Q}$	20,39, 40	17	19					
Q25	1S3 to 1$\overline{Q}$	32,13,4	9,24	12	22	10	35		
Q26	1S3 to 2$\overline{Q}$	32,13, 11	9,6	12	29				
Q27	3S3 to 3$\overline{Q}$	39,40, 19	17,14	20	36				
Q28	1S2 to 1$\overline{Q}$	32,24	12,23,9	13	22				
Q29	1S2 to 2$\overline{Q}$	32,6	12,7,9	13	29				
Q30	3S2 to 3$\overline{Q}$	39,14	20,15, 17	40	36				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q31	1S1 to 1$\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 50 pF	13,24	9,12,3	32	22	10	35	ns	15,17
Q32	1S1 to 2$\overline{Q}$		13,6	9,12,27	32	29	10	35		15,17
Q33	3S1 to 3$\overline{Q}$		40,14	20,17, 34	39	36	10	35		15,17
Q34	1E to 1$\overline{Q}$		32,23	12,13	9	22	8	33		15,16
Q35	1E to 2$\overline{Q}$		32,7	12,13	9	29		33		15,16
Q36	3E to 3$\overline{Q}$		39,15	20,40	17	36		33		15,16
Q37	1D0 to 1$\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	9,12, 13,32	2	22		36		15,17
Q38	1D1 to 1$\overline{Q}$		32	9,12,13	23					
Q39	1D2 to 1$\overline{Q}$		13	9,12,32	3					
Q40	1D3 to 1$\overline{Q}$		32,13	9,12	24					
Q41	1D4 to 1$\overline{Q}$		12	9,32,13	26					
Q42	1D5 to 1$\overline{Q}$		12,32	9,13	5					
Q43	1D6 to 1$\overline{Q}$		13,12	9,32	25					
Q44	1D7 to 1$\overline{Q}$		12,13, 32	9	4					
Q45	2D0 to 2$\overline{Q}$		Open	9,12, 13,32	28	29				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q46	2D1 to 2$\bar{Q}$	Propagation delay time (t_{PHL}), C_L = 150 pF	32	9,13,12	7	29	8	36	ns	15,17
Q47	2D2 to 2$\bar{Q}$		13	9,12,32	27					
Q48	2D3 to 2$\bar{Q}$		32,13	9,12	6					
Q49	2D4 to 2$\bar{Q}$		12	9,32,13	33					
Q50	2D5 to 2$\bar{Q}$		12,32	9,13	30					
Q51	2D6 to 2$\bar{Q}$		13,12	9,32	31					
Q52	2D7 to 2$\bar{Q}$		12,13, 32	9	11					
Q53	3D0 to 3$\bar{Q}$		Open	17,20, 39,40	35	36				
Q54	3D1 to 3$\bar{Q}$		39	17,20, 40	15					
Q55	3D2 to 3$\bar{Q}$		40	17,39, 20	34					
Q56	3D3 to 3$\bar{Q}$		39,40	17,20	14					
Q57	3D4 to 3$\bar{Q}$		20	17,39, 40	37					
Q58	3D5 to 3$\bar{Q}$		20,39	17,40	18					
Q59	3D6 to 3$\bar{Q}$		20,40	17,39	38					
Q60	3D7 to 3$\bar{Q}$		20,39, 40	17	19					

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q61	1S3 to $1\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 150 pF	32,13,4	9,24	12	22	15	48	ns	15,17
Q62	1S3 to $2\overline{Q}$		32,13, 11	9,6	12	29				
Q63	3S3 to $3\overline{Q}$		39,40, 19	17,14,	20	36				
Q64	1S2 to $\overline{1Q}$		32,24	12,23,9	13	22				
Q65	1S2 to $\overline{2Q}$		32 6	12,7,9	13	29				
Q66	3S2 to $\overline{3Q}$		39,14	20,15, 17	40	36				
Q67	1S1 to $1\overline{Q}$		13,24	9,12,3	32	22				
Q68	1S1 to $2\overline{Q}$		13,6	9,12,27	32	29				
Q69	3S1 to $3\overline{Q}$		40,14	20,17, 34	39	36				
Q70	1F to $1\overline{Q}$		32,23	12,13	9	22		44		15,16
Q71	1F to $2\overline{Q}$		32,7	12,13	9	29		44		
Q72	3F to $3\overline{Q}$		39,15	20,40	17	36		44		
Q73	1D0 to 1Q	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	9,12, 13,32	2	21	10	28		
Q74	1D1 to 1Q	Propagation delay time (t_{PHL}), C_L = 50 pF	32	9,13,12	23	21	10	28		

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q75	1D2 to 1Q Propagation delay time (t_{PHL}), C_L = 50 pF	13	9,12,32	3	21	10	28	ns	15,16
Q76	1D3 to 1Q	32,13	9,12	24					
Q77	1D4 to 1Q	12	9,32,13	26					
Q78	1D5 to 1Q	32,12	9,13	5					
Q79	1D6 to 1Q	12,13	9,32	25					
Q80	1D7 to 1Q	12,13 32	9	4					
Q81	2D0 to 2Q	Open	9,12 13,32	28	8				
Q82	2D1 to 2Q	32	9,13,12	7					
Q83	2D2 to 2Q	13	9,12,32	27					
Q84	2D3 to 2Q	32,13	9,12	6					
Q85	2D4 to 2Q	12	9,32,13	33					
Q86	2D5 to 2Q	32,12	9,13	30					
Q87	2D6 to 2Q	12,13	9,32	31					
Q88	2D7 to 2Q	12,13, 32	9	11					
Q89	3D0 to 3Q	Open	17,20, 39,40	35	16				
Q90	3D1 to 3Q	39	17,20, 40	15	16				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q91	3D2 to 3Q	Propagation delay time (t_{PHL}), C_L = 50 pF	40	17,20, 39	34	16	10	28	ns	15,16
Q92	3D3 to 3Q		39,40	17,20	14					
Q93	3D4 to 3Q		20	17,39, 40	37					
Q94	3D5 to 3Q		39,20	17,40	18					
Q95	3D6 to 3Q		20,40	17,39	38					
Q96	3D7 to 3Q		20,39, 40	17	19					
Q97	1S3 to 1Q		32,13,4	9,24	12	21		50		
Q98	1S3 to 2Q		32,13, 11	9,6	12	8				
Q99	3S3 to 3Q		39,40, 19	17,14	20	16				
Q100	1S2 to 1Q		32,24	12,23,9	13	21				
Q101	1S2 to 2Q		32,6	12,7,9	13	8				
Q102	3S2 to 3Q		39,14	20,15, 17	40	16				
Q103	1S1 to 1Q		13,24	9,12,3	32	21				
Q104	1S1 to 2Q		13,6	9,12,27	32	8				
Q105	3S1 to 3Q		40,14	20,17, 34	39	16				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q106	1E to 1Q	Propagation delay time (t_{PHL}), C_L = 50 pF	32,23	12,13	9	21	8	34	ns	15,17
Q107	1E to 2Q	Propagation delay time (t_{PHL}), C_L = 50 pF	32,7	12,13	9	8	8	34		15,17
Q108	3E to 3Q	Propagation delay time (t_{PHL}), C_L = 50 pF	33,15	20,40	17	16	8	34		15,17
Q109	1D0to 1Q	Propagation delay time (t_{PHL}), C_L = 150 pF	Open	9,12,13, 32	2	21	15	49		15,16
Q110	1D1 to 1Q		32	9,13,12	23					
Q111	1D2 to 1Q		13	9,12,32	3					
Q112	1D3 to 1Q		32,13	9,12	24					
Q113	1D4 to 1Q		12	9,32,13	26					
Q114	1D5 to 1Q		32,12,	9,13	5					
Q115	1D6 to 1Q		12,13	9,32	25					
Q116	1D7 to 1Q		12,13, 32	9	4					
Q117	2D0 to 2Q		Open	9,12, 13,32	28	8				
Q118	2D1 to 2Q		32	9,13,12	7	8				
Q119	2D2 to 2Q		13	9,12,32	27	8				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q120	2DF to 2Q Propagation delay time (t_{PHL}), C_L = 150 pF	32,13	9,12	6	8	15	49	ns	15,16
Q121	2D4 to 2Q	12	9,32,13	33					
Q122	2D5 to 2Q	32 12	9,13	30					
Q123	2D6 to 2Q	12,13	9,32	31					
Q124	2D7 to 2Q	12,13,32	9	11					
Q125	3D0 to 3Q	Open	17,20, 39,40	35	16				
Q126	3D1 to 3Q	39	17,20, 40	15					
Q127	3D2 to 3Q	40	17,20, 39	34					
Q128	3D3 to 3Q	39,40	17,20	14					
Q129	3D4 to 3Q	20	17,39, 40	37					
Q130	3D5 to 3Q	39,20	17,40	18					
Q131	3D6 to 3Q	20,40	17,39	38					
Q132	3D7 to 3Q	20 39, 40	17	19					
Q133	1S3 to 1Q	32,13,4	9,24	12	21	20	64		
Q134	1S3 to 2Q	32,13, 11	9,6	12	8	20	64		

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q135	3S3 to 3Q	Propagation delay time (t_{PHL}), C_L = 150 pF	39,40, 19	17,14	20	16	20	64	ns	15,16
Q136	1S2 to 1Q		32,24	12,23,9	13	21				
Q137	1S2 to 2Q		32,6	12,7,9	13	8				
Q138	3S2 to 3Q		39,14	20,15, 17	40	16				
Q139	1S1 to 1Q		13,24	9,12,3	32	21				
Q140	1S1 to 2Q		13,6	9,12,27	32	8				
Q141	3S1 to 3Q		40,14	20,17, 34	39	16				
Q142	1E to 1Q		32,23	12,13	9	21	15	50		15,17
Q143	1E to 2Q		32,7	12,13	9	8	15	50		15,17
Q144	3E to 3Q		39,15	20,40	17	16	15	50		15,17
Q145	1S3 to $1\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 50 pF	32,13,24	9,4	12	22	8	30		15,16
Q146	1S3 to $2\overline{Q}$		32,13,6	9,11	12	29				
Q147	3S3 to $3\overline{Q}$		39,40, 14	17,19	20	36				
Q148	1S2 to $1\overline{Q}$		32,23	12,24,9	13	22				
Q149	1S2 to $2\overline{Q}$		32,7	12,6,9	13	29				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q150	3S3 to $3\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 50 pF	39,15	20,14 17	40	35	8	30	ns	15,16
Q151	1S1 to $1\overline{Q}$		13,3	9,12,24	32	22				
Q152	1S1 to $2\overline{Q}$		13,27	9,12,6	32	29				
Q153	3S1 to $3\overline{Q}$		40,34	20,17 14	39	36				
Q154	1S3 to 1Q		32,13 31	9,4	12	21	10	35		15,17
Q155	1S3 to 2Q		32,13,6	9,11	12	8				
Q156	3S3 to 3Q		39,40, 14	17,19	20	16				
Q157	1S2 to 1Q		32,23	12,24,9	13	21				
Q158	1S2 to 2Q		32,7	12,6,9	13	8				
Q159	3S2 to 3Q		39,15	20,14, 17	40	16				
Q160	1S1 to 1Q		13,3	9,12,24	32	21				
Q161	1S1 to 2Q		13,27	9,12,6	32	8				
Q162	3S1 to 3Q		40,34	20,17, 14	39	16				
Q163	1S3 to $1\overline{Q}$	Propagation delay time (t_{PHL}), C_L = 150 pF	31,13, 24	9,4	12	22	15	41		15,16

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q164	1S3 to 2$\overline{Q}$ Propagation delay time (t_{PHL}), C_L = 150 pF	32,13,6	9,11	12	29	15	41	ns	15,16
Q165	3S3 to 3$\overline{Q}$	39,40, 14	17,19	20	36				
Q166	1S2 to 1$\overline{Q}$	32,23	12,24,9	13	22				
Q167	1S2 to 2$\overline{Q}$	32,7	12,6,9	13	29				
Q168	3S2 to 3$\overline{Q}$	39,15	20,14, 17	40	36				
Q169	1S1 to 1$\overline{Q}$	13,3	9,12,24	32	22				
Q170	1S1 to 2$\overline{Q}$	13,27	9,12,6	32	29				
Q171	3S1 to 3$\overline{Q}$	40,34	20,17, 14	39	36				
Q172	1S3 to 1Q	32,13, 24	9 4	12	21		48		15,17
Q173	1S3 to 2Q	32,13,6	9,11	12	8				
Q174	3S3 to 3Q	39,40, 14	17,19	20	16				
Q175	1S2 to 1Q	32,23	12,24,9	13	21				
Q176	1S2 to 2Q	32,7	12,6,9	13	8				
Q177	3S2 to 3Q	39,15	20,14, 17	40	16				
Q178	1S1 to 1Q	13,3	9,12,24	32	21				
Q179	1S1 to 2Q	13,27	9,12,6	32	8				
Q180	3S1 to 3Q	40,34	20,17, 14	39	16				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1D0 to 1$\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 50 pF	Open	9,12, 13,32	2	22	6	26	ns	15,17
R2	1D1 to 1$\overline{Q}$	32	9,13,12	23					
R3	1D2 to 1$\overline{Q}$	13	9,12,32	3					
R4	1D3 to 1$\overline{Q}$	32,13	9,12	24					
R5	1D4 to 1$\overline{Q}$	12	9,32,13	26					
R6	1D5 to 1$\overline{Q}$	32,12	9,13	5					
R7	1D6 to 1$\overline{Q}$	12,13	9,32	25					
R8	1D7 to 1$\overline{Q}$	12,13, 32	9	4					
R9	2D0 to 2$\overline{Q}$	Open	9,12, 13,32	28	29				
R10	2D1 to 2$\overline{Q}$	32	9,13,12	7					
R11	2D2 to 2$\overline{Q}$	13	9,12,32	27					
R12	2D3 to 2$\overline{Q}$	32,13	9,12	6					
R13	2D4 to 2$\overline{Q}$	12	9,32,13	33					
R14	2D5 to 2$\overline{Q}$	32,12	9,13	30					
R15	2D6 to 2$\overline{Q}$	12 13	9,32	31					
R16	2D7 to 2$\overline{Q}$	12,13, 32	9	11					

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R17	3D0 to 3$\bar{Q}$ Propagation delay time (t_{PLH}), C_L = 50 pF	Open	17,20, 39,40	35	36	6	26	ns	15,17
R18	3D1 to 3$\bar{Q}$	39	17,20, 40	15					
R19	3D2 to 3$\bar{Q}$	40	17,20. 39	31					
R20	3D3 to 3$\bar{Q}$	39,40	17,20,	14					
R21	3D4 to 3$\bar{Q}$	20	17,39, 40	37					
R22	3D5 to 3$\bar{Q}$	39,20	17,40	18					
R23	3D6 to 3$\bar{Q}$	20,40	17,39	38					
R24	3D7 to 3$\bar{Q}$	20,39 40	17	19					
R25	1S3 to 1$\bar{Q}$	32,13,4	9,24	12	22	10	45		
R26	1S3 to 2$\bar{Q}$	32,13, 11	9,6	12	29				
R27	3S3 to 3$\bar{Q}$	39,40, 19	17,14	20	36				
R28	1S2 to 1$\bar{Q}$	32,24	12,23,9	13	22				
R29	1S2 to 2$\bar{Q}$	32,6	12,7,9	13	29				
R30	3S2 to 3$\bar{Q}$	39,14	20,15, 17	40	36				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R31	1S1 to 1$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 50 pF	13,24	9,12,3	32	22	10	45	ns	15,17
R32	1S1 to 2$\overline{Q}$		13,6	9,12,27	32	29	10	45		15,17
R33	3S1 to 3$\overline{Q}$		40,14	20,17, 34	39	36	10	45		15,17
R34	1E to 1$\overline{Q}$		32,23	12,13	9	22	8	29		15,16
R35	1E to 2$\overline{Q}$		32,7	12,13	9	29	8	29		15,16
R36	3E to 3$\overline{Q}$		39,15	20,40	17	36	8	29		15,16
R37	1D0 to 1$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	9,12, 13,32	2	22	10	40		15,17
R38	1D1 to 1$\overline{Q}$		32	9,12,13	23					
R39	1D2 to 1$\overline{Q}$		13	9,12,32	3					
R40	1D3 to 1$\overline{Q}$		32,13	9,12	24					
R41	1D4 to 1$\overline{Q}$		12	9,32,13	26					
R42	1D5 to 1$\overline{Q}$		12,32	9,13	5					
R43	1D6 to 1$\overline{Q}$		12,13	9,32	25					
R44	1D7 to 1$\overline{Q}$		12,13, 32	9	4					
R45	2D0 to 2$\overline{Q}$		Open	9,12, 13,32	28	29				

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TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R46	2D1 to $2\overline{Q}$ Propagation delay time (t_{PLH}), C_L 150 pF	32	9,13,12	7	29	10	40	ns	15,17
R47	2D2 to $2\overline{Q}$	13	9,12,32	27					
R48	2D3 to $2\overline{Q}$	32,13	9,12	6					
R49	2D4 to $2\overline{Q}$	12	9,32,13	33					
R50	2D5 to $2\overline{Q}$	12,32	9,13	30					
R51	2D6 to $2\overline{Q}$	12,13	9,32	31					
R52	2D7 to $2\overline{Q}$	12,13, 32	9	11					
R53	3D0 to $3\overline{Q}$	Open	17,20, 39,40	35	36				
R54	3D1 to $3\overline{Q}$	39	17,20, 40	15					
R55	3D2 to $3\overline{Q}$	40	17,20, 39	34					
R56	3D3 to $3\overline{Q}$	39,40	17,20	14					
R57	3D4 to $3\overline{Q}$	20	17,39, 40	37					
R58	3D5 to $3\overline{Q}$	39,20	17,40	18					
R59	3D6 to $3\overline{Q}$	20,40	17,39	38					
R60	3D7 to $3\overline{Q}$	20,39, 40	17	19					

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R61	1S3 to 1$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 150 pF	32,13,4	9,24	12	22	15	56	ns	15,17
R62	1S3 to 2$\overline{Q}$		32,13, 11	9,6	12	29				
R63	3S3 to 3$\overline{Q}$		39,40, 19	17,14	20	36				
R64	1S2 to 1$\overline{Q}$		32,24	12,23,9	13	22				
R65	1S2 to 2$\overline{Q}$		32,6	12,7,9	13	29				
R66	3S2 to 3$\overline{Q}$		39,14	20,15, 17	40	36				
R67	1S1 to 1$\overline{Q}$		13,24	9,12,3	32	22				
R68	1S1 to 2$\overline{Q}$		13,6	9,12,27	32	29				
R69	3S1 to 3$\overline{Q}$		40,14	20,17, 34	39	36				
R70	1E to 1$\overline{Q}$		32,23	12,13	9	22		43		15 16
R71	1E to 2$\overline{Q}$		32,7	12,13	9	29		43		
R72	3E to 3$\overline{Q}$		39,15	20,40	17	36		43		
R73	1D0 to 1Q	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	9,12, 13,32	2	21	10	33		
R74	1D1 to 1Q	Propagation delay time (t_{PLH}), C_L = 50 pF	32	9,13,12	23	21	10	33		

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R75	1D2 to 1Q Propagation delay time (t_{PLH}), C_L = 50 pF	13	9,12,32	3	21	10	33	ns	15,16
R76	1D3 to 1Q	32,13	9,12	24					
R77	1D4 to 1Q	12	9,32,13	26					
R78	1D5 to 1Q	32,12	9,13	5					
R79	1D6 to 1Q	12,13	9,32	25					
R80	1D7 to 1Q	12,13, 32	9	4					
R81	2D0 to 2Q	Open	9,12, 13,32	28	8				
R82	2D1 to 2Q	32	9,13,12	7					
R83	2D2 to 2Q	13	9,12,32	27					
R84	2D3 to 2Q	32,13	9,12	6					
R85	2D4 to 2Q	12	9,32,13	33					
R86	2D5 to 2Q	32,12	9,13	30					
R87	2D6 to 2Q	12,13	9,32	31					
R88	2D7 to 2Q	12,13, 32	9	11					
R89	3D0 to 3Q	Open	17,20, 39,40	35	16				
R90	3D1 to 3Q	39	17,20, 40	15	16				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P91	3D7 to 3Q	Propagation delay time (t_{PLH}), C_L = 50 pF	40	17,20,39	34	16	10	33	ns	15,16
R92	3D3 to 3Q		39,40	17,20	14					
R93	3D4 to 3Q		20	17,39, 40	37					
P94	3D5 to 3Q		39,20	17,40	18					
R95	3D6 to 3Q		20,40	17,39	38					
R96	3D7 to 3Q		20,39, 40	17	19					
R97	1S3 to 1Q		32,13,4	9,24	12	21	15	51		
R98	1S3 to 2Q		32,13, 11	9,6	12	8				
R99	3S3 to 3Q		39,40, 19	17,14	20	16				
R100	1S2 to 1Q		32,24	12,23,9	13	21				
R101	1S2 to 2Q		32,6	12,7,9	13	8				
R102	3S2 to 3Q		39,14	20,15, 17	40	16				
R103	1S1 to 1Q		13,24	9,12,3	32	21				
R104	1S1 to 2Q		13,6	9,12,27	32	8				
R105	3S1 to 3Q		40,14	20,17 34	39	16				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R106	1F to 1Q	Propagation delay time (t_{PLH}), C_L = 50 pF	32,33	12,13	9	21	15	43	ns	15,17
R107	1F to 2Q	Propagation delay time (t_{PLH}), C_L = 50 pF	32,33	12,13	9	8		43		15,17
R108	3F to 3Q	Propagation delay time (t_{PLH}), C_L = 50 pF	39,15	20,40	17	16		43		15,17
R109	1D0 to 1Q	Propagation delay time (t_{PLH}), C_L = 150 pF	Open	9,12, 13,32	7	21		48		15,16
R110	1D1 to 1Q		33	9,13,12	23					
R111	1D2 to 1Q		13	9,12,32	3					
R112	1D3 to 1Q		32,13	9,12	24					
R113	1D4 to 1Q		12	9,32,13	26					
R114	1D5 to 1Q		32,12	9,13	5					
R115	1D6 to 1Q		12,13	9,32	25					
R116	1D7 to 1Q		12,13, 32	9	4					
R117	2D0 to 2Q		Open	9,12 13,32	28	8				
R118	2D1 to 2Q		33	9,13,12	7					
R119	2D2 to 2Q		13	9,12,32	27	8				

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter		Connect				Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R120	2D3 to 2Q	Propagation delay time (t_{PLH}), C_L = 150 pF	32,13	9,12	6	8	15	48	ns	15,16
R121	2D4 to 2Q		12	9,32,13	33					
R122	2D5 to 2Q		32,12	9,13	30					
R123	2D6 to 2Q		12,13	9,32	31					
R124	2D7 to 2Q		12,13, 31	9	11					
R125	3D0 to 3Q		Open	17 20, 39,40	35	16				
R126	3D1 to 3Q		39	17,20, 40	15					
R127	3D2 to 3Q		40	17,20, 39	34					
R128	3D3 to 3Q		39,40	17,20	14					
R129	3D4 to 3Q		20	17,39, 40	37					
R130	3D5 to 3Q		39,20	17,40	18					
R131	3D6 to 3Q		20,40	17,39	38					
R132	3D7 to 3Q		20,39, 40	17	19					
R133	1S3 to 1Q		32,13,4	9,24	12	21	20	62		
R134	1S3 to 2Q		32,13, 11	9,6	12	8	20	62		

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R135	3S3 to 3Q	Propagation delay time (t_{PLH}), C_L = 150 pF	39,40, 19	17,14	20	16	20	62	ns	15,16
R136	1S2 to 1Q		32,24	12,23,9	13	21				
R137	1S2 to 2Q		32,6	12,7,9	13	8				
R138	3S2 to 3Q		39,14	20,15, 17	40	16				
R139	1S1 to 1Q		33,24	9,12,3	32	21				
R140	1S1 to 2Q		33,6	9,12,27	32	8				
R141	3S1 to 3Q		40,14	20,17, 34	39	16				
R142	1E to 1Q		32,23	12,13	9	21		54		15,17
R143	1E to 2Q		32,7	12,13	9	8		54		15,17
R144	3E to 3Q		39,15	20,40	17	16		54		15,17
R145	1S3 to 1$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 50 pF	32,13, 24	9,4	12	22	6	30		15,16
R146	1S3 to 2$\overline{Q}$		32,13,6	9,11	12	29				
R147	3S3 to 3$\overline{Q}$		39,40 11	17,19	20	36				
R148	1S2 to 1$\overline{Q}$		32,23	12,24,9	13	22				
R149	1S2 to 2$\overline{Q}$		31,7	12,6,9	13	29				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

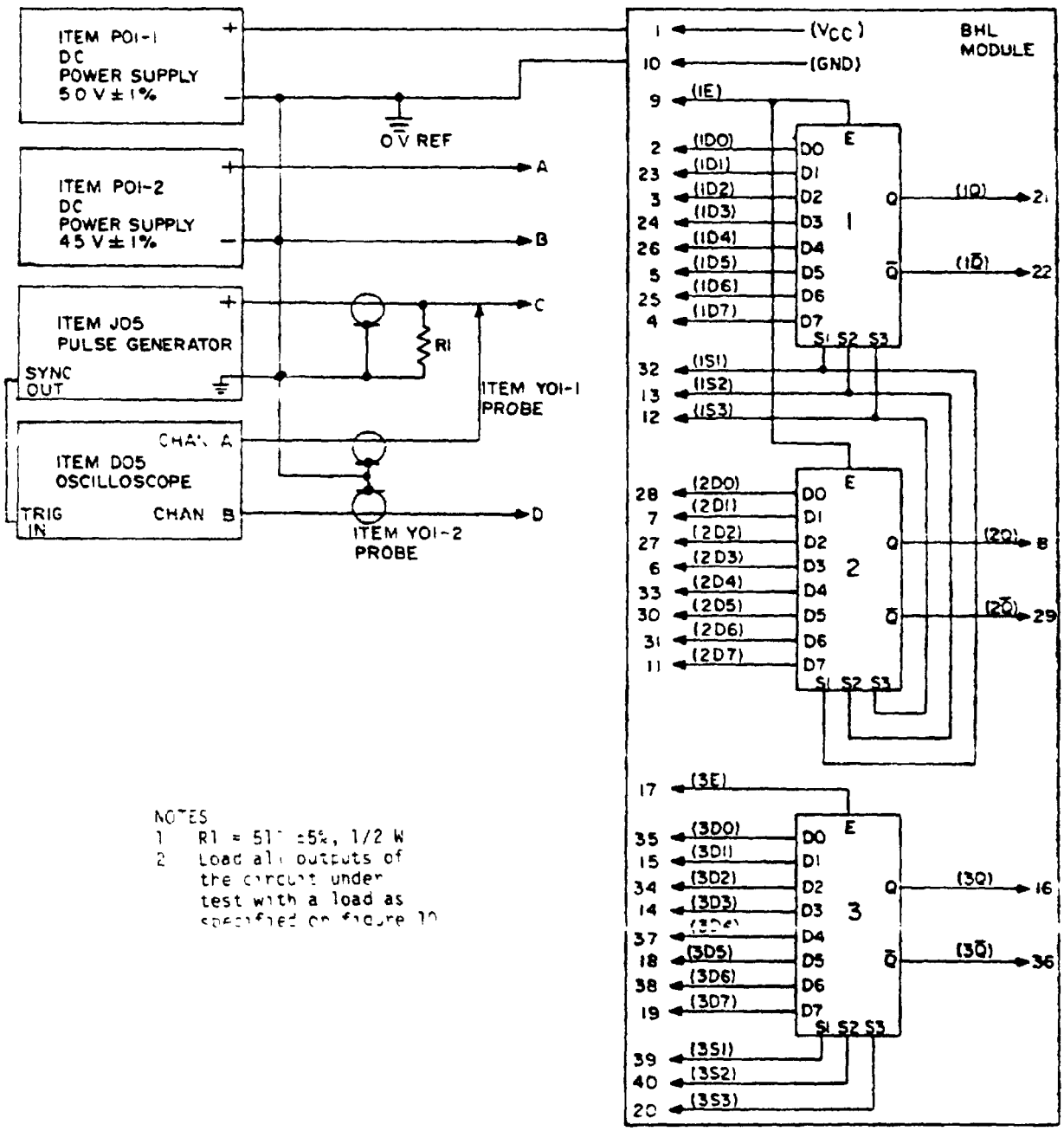
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R150	3S2 to 3$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 50 pF	39,15	20,14, 17	40	36	6	30	ns	15,16
R151	1S1 to 1$\overline{Q}$		13,3	9,12,24	32	22				
R152	1S1 to 2$\overline{Q}$		13,27	9,12,6	32	29				
R153	3S1 to 3$\overline{Q}$		40,34	20,17, 14	39	36				
R154	1S3 to 1Q		32,13, 24	9,4	12	21	15	44		15,17
R155	1S3 to 2Q		32,13,6	9,11	12	8				
R156	3S3 to 3Q		39,40, 14	17,19	20	16				
R157	1S2 to 1Q		32,23	12,24,9	13	21				
R158	1S2 to 2Q		32,7	12,6,9	13	8				
R159	3S2 to 3Q		39,15	20,14, 17	40	16				
R160	1S1 to 1Q		13,3	9,12,24	32	21				
R161	1S1 to 2Q		13,27	9,12,6	32	8				
R162	3S1 to 3Q		40,34	20,17, 14	39	16				
R163	1S3 to 1$\overline{Q}$	Propagation delay time (t_{PLH}), C_L = 100 pF	31,13, 24	9,4	12	22		43		15,16

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	U to pin	Min	Max		
R164	1S3 to 2$\overline{Q}$ Propagation delay time (t_{PLH}), C_L = 150 pF	32,13,6	9,11	12	29	15	43	ns	15,16
R165	3S3 to 3$\overline{Q}$	39,40, 14	17,19	20	36				
R166	1S2 to 1$\overline{Q}$	32,23	12,24,9	13	22				
R167	1S2 to 2$\overline{Q}$	32,7	12,6,9	13	29				
R168	3S2 to 3$\overline{Q}$	39,15	20,14, 17	40	36				
R169	1S1 to 1$\overline{Q}$	13,3	9,12,24	32	22				
R170	1S1 to 2$\overline{Q}$	13,27	9,12,6	32	29				
R171	3S1 to 3$\overline{Q}$	40,34	20,17, 14	39	36				
R172	1S3 to 1Q	32,13, 24	9,4	12	21	20	56		15,17
R173	1S3 to 2Q	32,13,6	9,11	12	8				
R174	3S3 to 3Q	39,40, 14	17,19	20	16				
R175	1S2 to 1Q	32,23	12,24,9	13	21				
R176	1S2 to 2Q	32,7	12,6,9	13	8				
R177	3S2 to 3Q	39,15	20,14, 17	40	16				
R178	1S1 to 1Q	13,3	9,12,24	32	21				
R179	1S1 to 2Q	13,27	9,12,6	32	8				
R180	3S1 to 3Q	40,34	20,17, 14	39	16				

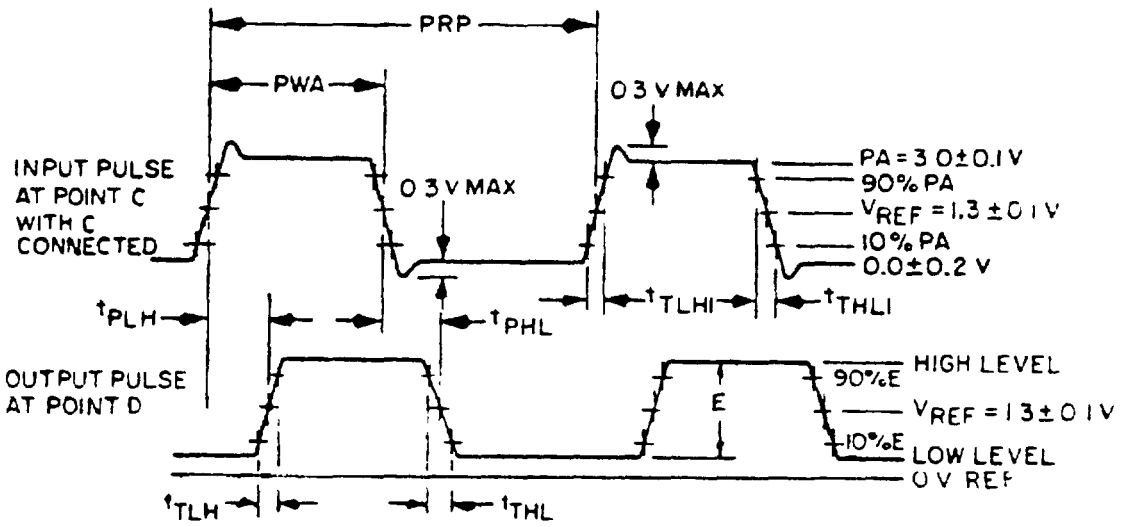
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NOTES
1. R1 = 51 ±5%, 1/2 W
2. Load all outputs of the circuit under test with a load as specified on figure 10

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIV and XV

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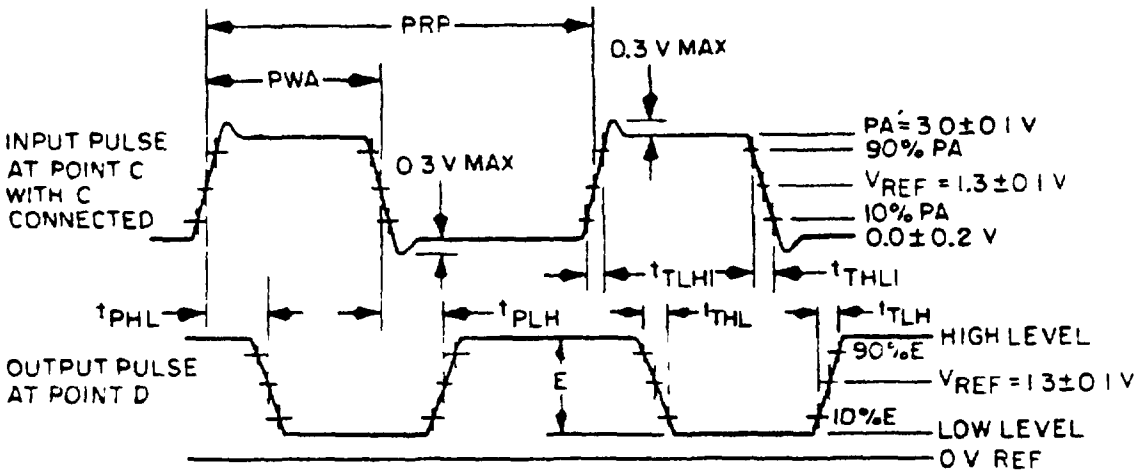


NOTES

1. Input pulse parameters
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
2. Before making the measurements specified in tables XIV and XV, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between probes Y01-1 and Y01-2 and output pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIV and XV.

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NOTES
1. Input pulse parameters
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns
   PWA = 500 ±10 ns
   PRP = 1,000 ±50 ns
2. Before making the measurements specified in tables XIV and XV, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between probes YO1-1 and YO1-2 and output pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIII and XV.

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TABLE XVI Output crosstalk (low-level) test.

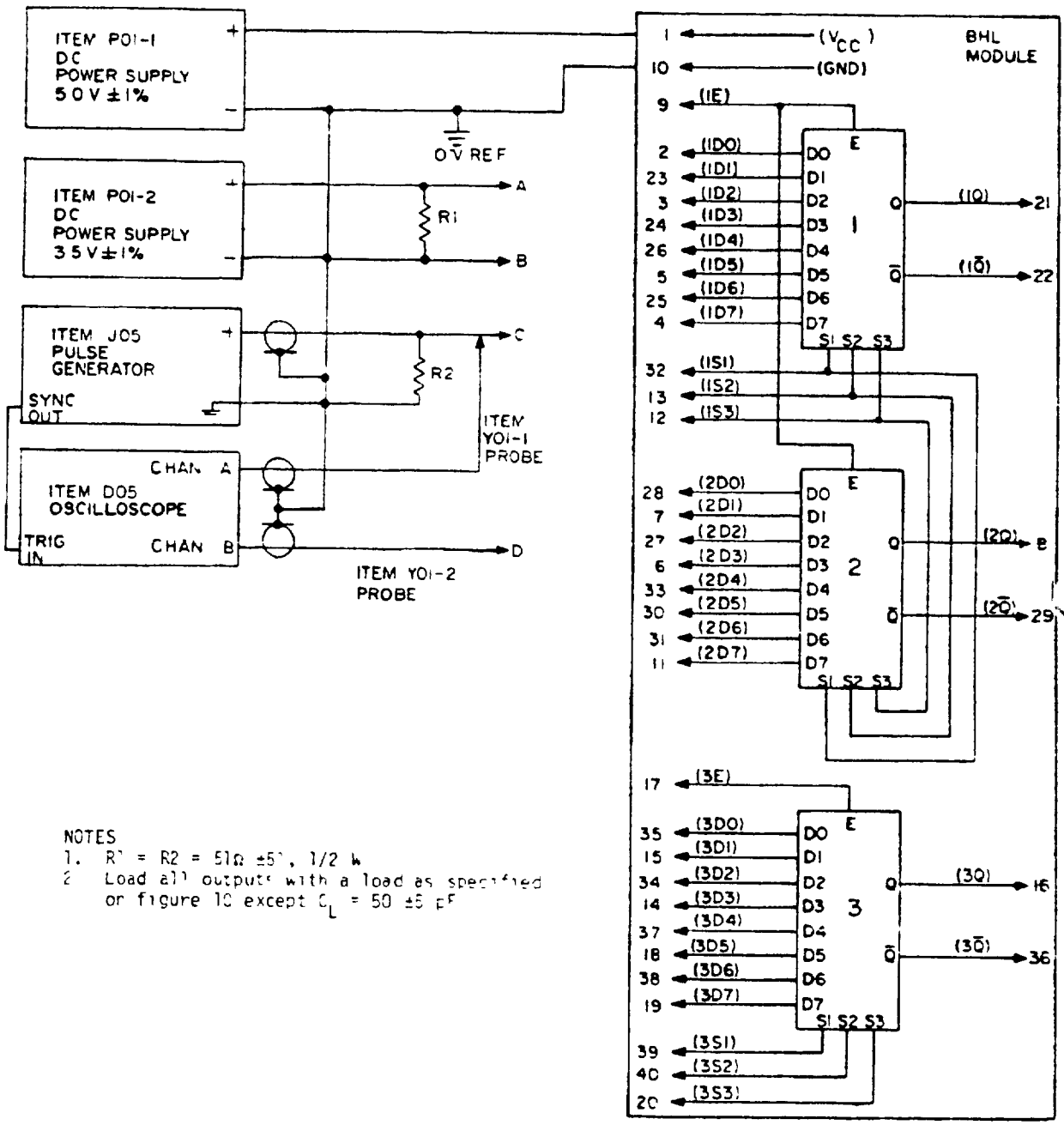
Test condition		Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Q	Output crosstalk, low level	32,13, 39,40	9,17, 12,20, 24	6,14	21	0	400	mV	18,19
S2	1$\bar{Q}$		32,13, 39,40, 24	9,17, 12,20	6,14	22				
S3	2Q		32,13, 39,40,	9,17, 12,20,6	24,14	8				
S4	2$\bar{Q}$		32,13, 39,40,6	9,17, 12,20	24,14	29				
S5	3Q		32,13, 39,40	9,17, 12,20, 14	24,6	16				
S6	3$\bar{Q}$		32,13, 39,40, 14	9,17, 12,20	24,6	36				

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TABLE XVII Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1Q	Output crosstalk, high level	32,13, 39,40 21	9,17, 12,20	6,14	21	0	400	mV	18,19
T2	$1\overline{Q}$	↓	32,13, 39,40	9,17, 12,24, 20	6,14	22	↓	↓	↓	↓
T3	2Q	↓	32,13, 39,40,6	9,17, 12,20	24,14	8	↓	↓	↓	↓
T4	$2\overline{Q}$	↓	32,13, 39,40	9,17, 12,6,20	24,14	29	↓	↓	↓	↓
T5	3Q	↓	32,13, 39,40 14	9,17, 12,20	24,6	16	↓	↓	↓	↓
T6	$3\overline{Q}$	↓	32,13, 39,40	9,17, 12,14, 20	24,6	36	↓	↓	↓	↓

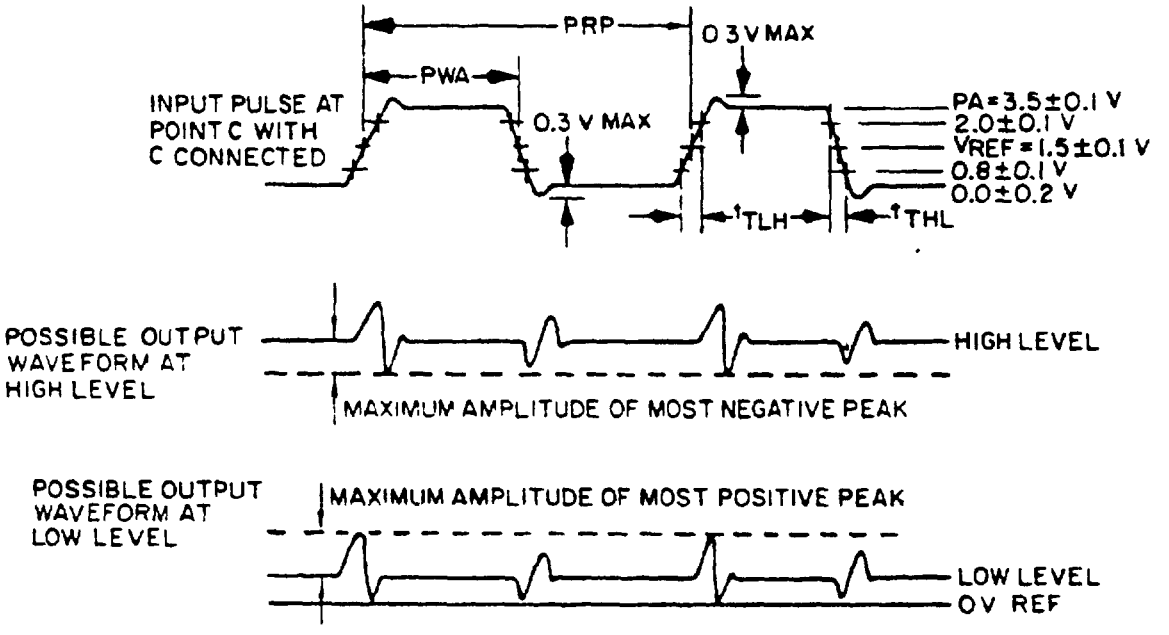
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. Load all outputs with a load as specified on figure 10 except C_L = 50 ±5 pF

* FIGURE 18. Output crosstalk (low-level and high-level) test circuit for tables XII and XIII.

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NOTES
1. Input pulse parameters
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin)
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
2. The lead length of electrical conductors between resistors R1 and R2 and input pins, between probes YO1-1 and YO1-2 and input and output pins, between load circuits and output pins, and between ground leads of probes YO1-1 and YO1-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm)

* FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for tables X.I and XII.

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TABLE XVIII Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	20,21

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	22

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TABLE XX. Input capacitance test.

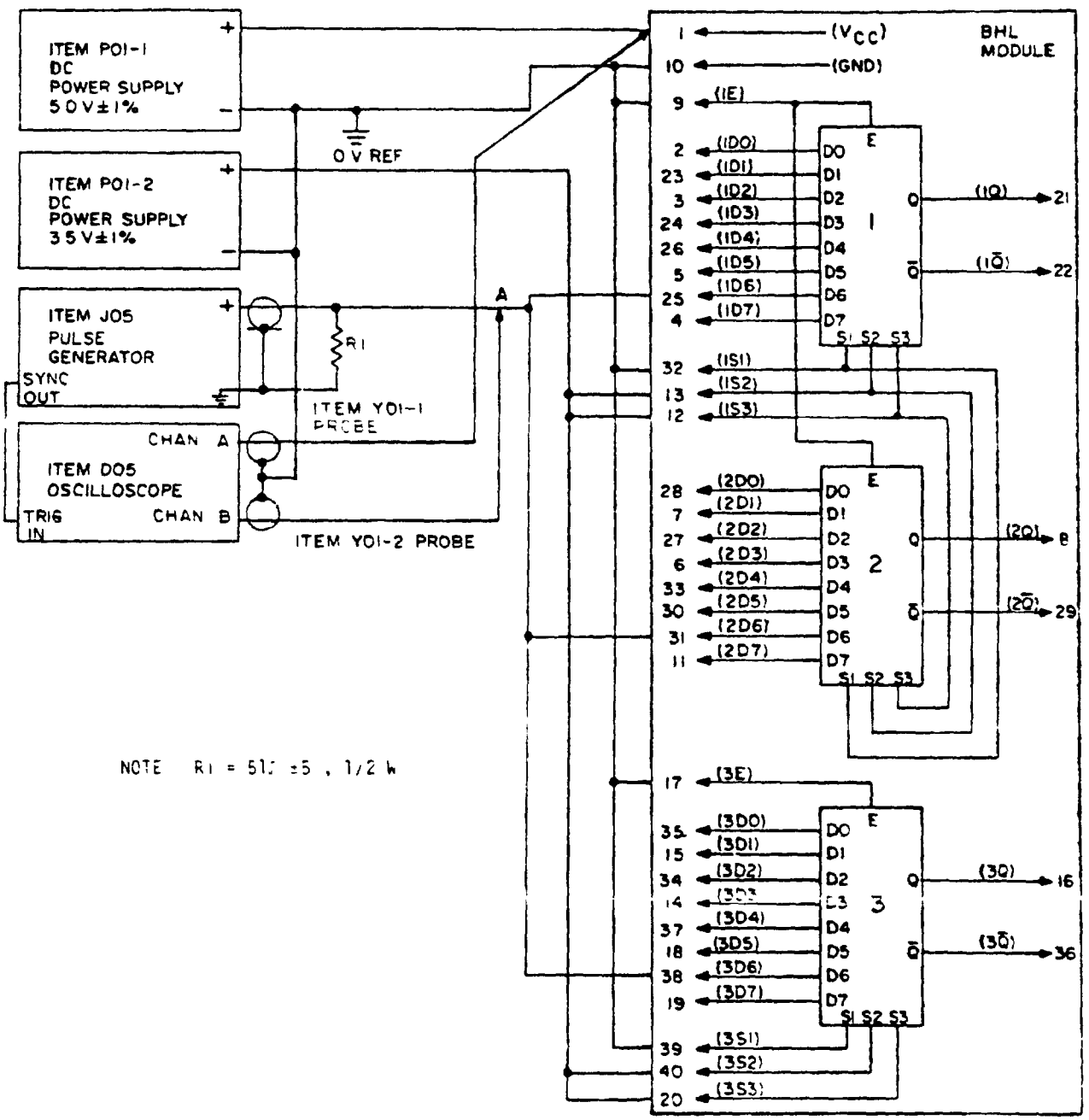
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	1D0 Input capacitance	2	Open	2	15	pF	22
W2	1D1	23					
W3	1D2	3					
W4	1D3	24					
W5	1D4	26					
W6	1D5	5					
W7	1D6	25					
W8	1D7	4					
W9	2D0	28					
W10	2D1	7					
W11	2D2	27					
W12	2D3	6					
W13	2D4	33					
W14	2D5	30					
W15	2D6	31					
W16	2D7	11					
W17	3D0	35					
W18	3D1	15					
W19	3D2	34					

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W20	3D3	Input capacitance	14	Open	2	15	pF	22
W21	3D4		37					
W22	3D5		18					
W23	3D6		38					
W24	3D7		19					
W25	3E		17					
W26	3S1		39					
W27	3S2		40					
W28	3S3		20					
W29	1E		9			30		
W30	1S1		32					
W31	1S2		13					
W32	1S3		12					

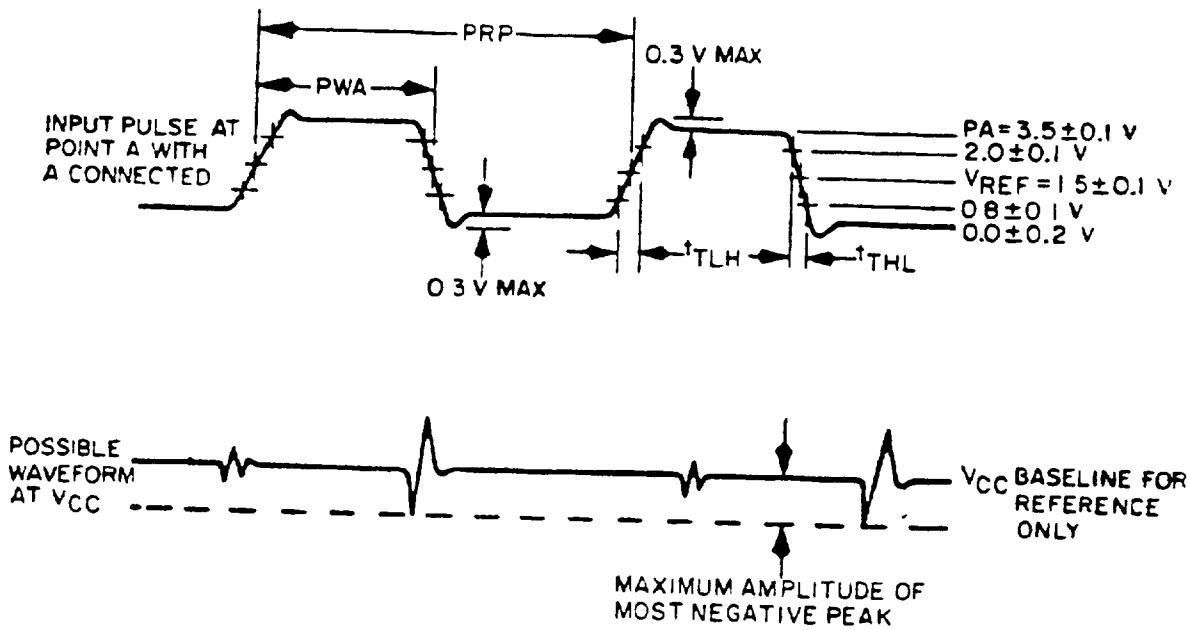
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NOTE R1 = 51Ω ±5 , 1/2 W

*FIGURE 20 Power supply transient amplitude test circuit for table XIII

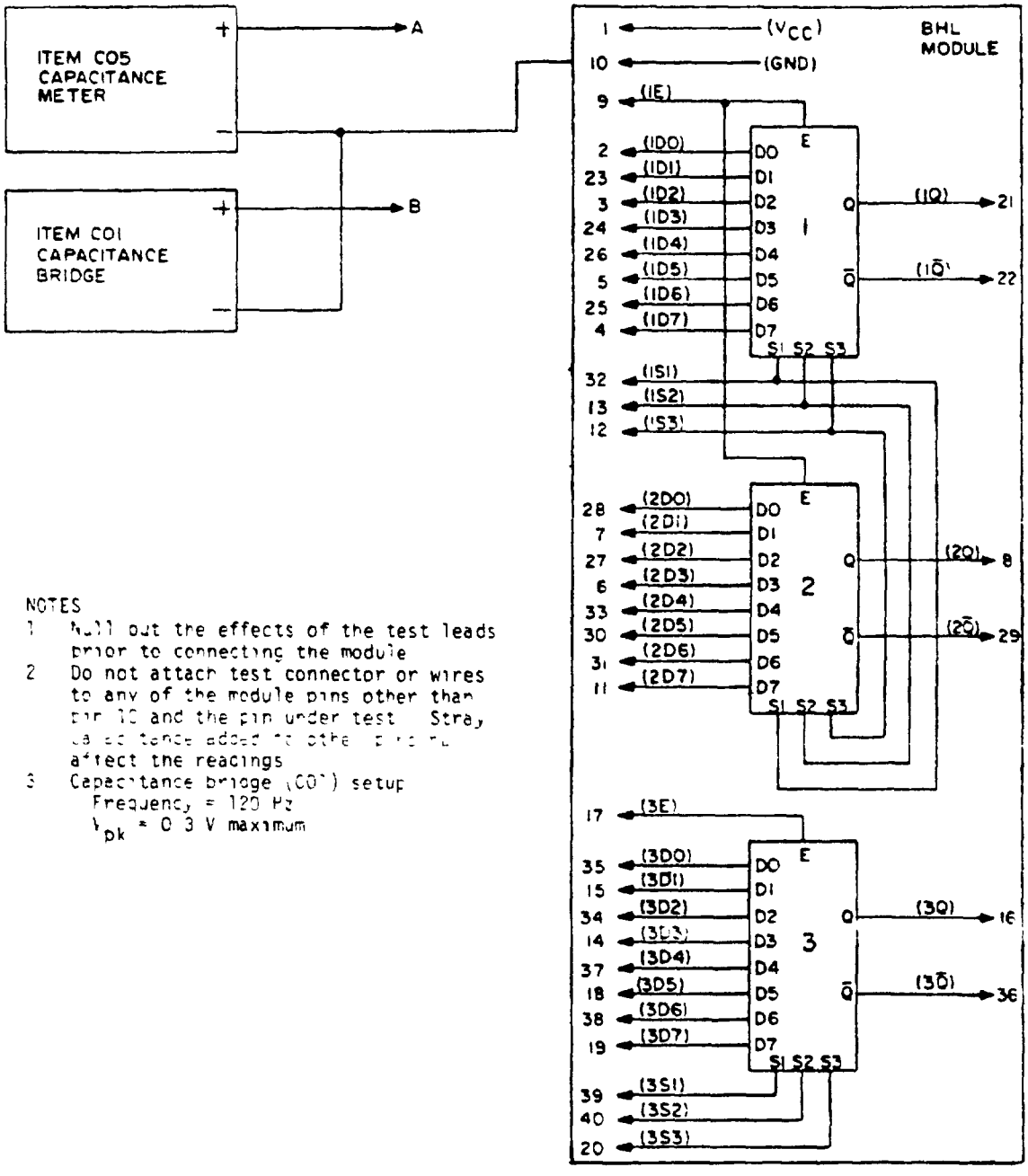
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NOTE Input pulse parameters:
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin)
PWA = 500 ±50 ns
PRP = 1,000 ±100 ns

*FIGURE 21 Power supply transient amplitude test circuit waveforms for table XVIII

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NOTES
1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3. Capacitance bridge (C01) setup
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum

*FIGURE 22. Power supply filter and input capacitance test circuit for tables XII and XX

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